Supported Morita Descent for Semistable Curves

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Abstract

We study geometric rational nearby-cycle modules on proper nodal curves over a fixed trait. A geometric constructibility profile specifies marked components, unipotent chain blocks and a finite system of supports. For actual integer-cut modifications we prove that these conditions select equivalent coefficient categories in every presentation, without choosing a coarse anchor. The comparisons retain arbitrary exterior attachments, horizontal and loop monodromy, localization boundaries and geometric Verdier pairings. On a source generated by support graphs and coefficient rectangles, an actual one-sided kernel counit gives units, strong convolution and a multi-object Morita realization. The same method handles finite etale support spans in existing presentations of a fixed cover diagram. A triangular cubic supplies continuous loop holonomy, a nonzero boundary, and both rectangular and coupled noninvertible kernels. An off-diagonal calculation shows why arbitrary independent contraction of the endpoints of a coupled kernel fails. The results concern the specified nearby-module support system; general log/root sources and higher-dimensional modifications are outside their scope.

1 Introduction

Fix a complete strictly henselian discrete valuation ring R with algebraically closed residue field k, uniformizer t, and $\ell \neq \operatorname{char} k$. Fix a completed algebraic closure of its fraction field and compatible base roots. We begin with proper nodal models of a smooth generic curve, with local equations $xy = t^m$, $m \geq 1$. The intended modifications are actual integer-cut toroidal modifications over this fixed trait and their composites. Within one connected integer-cut component, Section 5 constructs the common refinement and proves descent; arbitrary rational subdivisions are not included.

A support profile specifies finite diagrams of actual locally closed special-fibre supports and open/closed decompositions. Presentations are to be compared on complete inverse-image supports in common refinements. No coarse model is fixed for the entire theory. A new fine support need not represent the same profile as an old support. The initial kernel products are over k; a relative kernel theory over R is a separate extension.

The invariant sought is a diagram of categories and operations,

$$(\mathcal{C}_M(S), \mathcal{C}_M^D(S), j^*, j_!, Rj_*, i^*, i^!, i_*, D, \mathrm{ev}, \mathrm{Tr}),$$

restricted to operations for which preservation and finiteness are proved. The notation does not assume that all six operations preserve one small coefficient category.

1.1 The two main results

The coefficient algebra on a model M is $H_M = R\Psi_M \mathbf{Q}_\ell$. Its restriction to a smooth stratum is the constant algebra; its node stalk is the cochain algebra of the geometric vanishing annulus. We retain sheaf modules and their attaching maps, rather than replacing the whole curve by one cohomology algebra.

Theorem A: coefficients with supports. A profile marks geometric divisorial valuations on the canonical finest model. Unmarked components form rational chains; their coefficients are unipotent with invertible internal branch maps. Marked open strata have arbitrary finite-rank lisse coefficients, and all attachments between pieces are retained. Supports are specified unions and differences of whole pieces. Theorem 5.6 compares the resulting full coefficient categories for every presentation in one integer-cut component, together with their support diagrams and Verdier partners. The profile is defined geometrically before the descent theorem; Lemma 5.5 proves that all presentations select the same objects on the finest model.

The two coefficient types have distinct roles. Ordinary modules are transported by $F_u = H_{M'} \otimes^L_{u^* H_M} u^*(-)$, with inverse $G_u = Ru_*$ on the selected image. Their supported partners are obtained by geometric Verdier duality D and transported by $F^D_u = DF_uD$. Restriction and extension by zero use the ordinary type; their Verdier mates use the supported type. The failed ordinary Rj_* comparison in Example 4.5 explains why this distinction is necessary.

Theorem B: geometric kernel comparisons. The initial kernel source is generated by support graphs and rectangles of these coefficients, with all actual kernel maps. For a model change, Theorem 6.4 proves the actual counit

$$U_b \star (V_b \star P \star U_a) \longrightarrow P \star U_a$$

is invertible in the kernel category. Inserting this map proves strong convolution, and the diagonal gives the regular unit. Theorem 6.5 then gives the Morita realization and its geometric Verdier mates. It does not infer kernel equivalences from equivalences of their actions.

Section 7 keeps the same curve setting and distinguishes two extensions of B. Rectangles allow independently chosen presentations of several curves. Finite etale support spans allow coupled kernels, but the fixed generic cover diagram must have actual finite etale model presentations. Normalization uniqueness proves that their modification squares are Cartesian. The source is independently specified by those geometric models; existence on every possible coarse model is not a theorem here.

1.2 Proof structure and inputs

Sections 2–3 identify the local cochain source, compare coordinate torsors, and reconstruct actual modules by recollement, including mapping spectra. Section 4 proves one modification at a time: the exceptional chain is controlled by its unit, while the exterior attaching maps are compared by an ambient adjunction. Section 5 constructs the finest model and proves the common image needed for A. Section 6 proves boundedness and the actual counit before using kernel or Verdier formalism. Section 8 constructs the cyclic test curve first, then its rectangular and finite-cover kernels. The examples are applications, not inputs to the descent proofs.

The local node comparison and rational unipotent interface use [1]; the integer-cut fibre reconstruction uses [2]; the ambient enhanced kernel operations use [3]. The results invoked from these papers are named in the input table in Section 9, with archived source versions and the hypotheses used here. The new common-image, support-preservation and enlarged-source kernel-counit arguments are proved here. All products of special-fibre supports are over k.

1.3 Related work

Haine–Porta–Teyssier prove that coarsening an exodromic stratification localizes its exit-path infinity-category at the arrows made invertible by the coarsening [4, Theorem 0.3.1(3)]. This is a general precedent for the localization mechanism. We do not verify the hypotheses needed to apply that theorem directly to our enhanced etale nearby-algebra module categories. Instead, the work here identifies the actual coefficient source, the image of each exceptional chain and its compatibility with the specified support operations. Curry's subdivision theorem identifies cellular sheaf cohomology before and after pullback to a subdivision [5, Theorem 6.12]. Relative to that particular invariance statement, our comparison retains full module mapping spectra and actual exterior attachments, as well as localization boundaries and Verdier pairings. We do not claim subdivision invariance itself as a new formal mechanism; these geometric and supported checks are the content required in the present coefficient setting.

2 Coordinate changes and actual local coefficients

Put $Q = \mathbf{Q}_\ell$ and $\Lambda_r = \mathbf{Z}/\ell^r$. All nearby cycles in this section are geometric: no extra base-inertia representation is added. Let M/R be one of the actual nodal models above, put $Y = M_s$, and let $i : Z \hookrightarrow Y$ be its finite node locus and $j : U \hookrightarrow Y$ its smooth complement. The algebra

$$H = R\Psi_M Q, \qquad j^*H \simeq Q_U, \qquad A_q = i_q^* H \quad (q \in Z)$$

is defined by actual nearby cycles and their lax monoidal structure. The smooth restriction uses the actual specialization unit. These nearby algebras are used with their commutative enhanced structures. Ordinary coefficients are right modules; their Verdier partners are right modules over the opposite algebra, equivalently left modules over the original algebra. A kernel from a to b is left H_a and right H_b. When tensoring two ordinary coefficients, or forming a rectangle, commutativity supplies the required left action with the usual symmetry signs. Verdier duality transposes actions; it does not assert a scalar tensor-dual formula for an arbitrary constructible coefficient. We use the stable categories of sheaf modules over these algebras, with underlying bounded constructible Q-complexes when specified. No model contraction enters these definitions.

2.1 The unit torsor must be retained

For an invertible function f on a scheme or analytic domain on which ℓ is invertible, write

$$T_n(f) = \{z : z^{\ell^n} = f\}, \qquad G_n = \mu_{\ell^n}.$$

This is the actual right G_n-torsor. For a node coordinate x these torsors are taken on the geometric generic locus, where x is invertible; a unit torsor $T_n(a)$ extends etale over the chart. Write $T_n(f) \wedge^{G_n} T_n(g)$ for the quotient of their product by $(s,t) \mapsto (s\zeta, t\zeta^{-1})$. Its residual G_n-action is multiplication in either factor.

Lemma 2.1 (Unit-coordinate torsors and their descent cells)**.** *On an overlap where $x' = ax$ with a an invertible regular function, there is a canonical torsor isomorphism*

$$m_{a,x} : T_n(a) \wedge^{G_n} T_n(x) \xrightarrow{\sim} T_n(x'), \qquad [s,z] \longmapsto sz.$$

These maps commute with $T_{n+1}(f) \to T_n(f)$ and with restriction. For $x'' = bx'$, their two composites agree through the canonical contracted-product associator and $m_{b,a}$. In a node chart $xy = t^m$, compatible base roots $t_n^{\ell^n} = t$ give

$$T_n(x) \longrightarrow T_n(y), \qquad z \longmapsto t_n^m/z,$$

where the group action is inverted. These comparisons, including branch interchange and unit changes together, satisfy all the cells induced by multiplication and inversion of the torsors.

Proof. Multiplication is invariant under the displayed antidiagonal action. After an etale cover trivializing the two torsors, it is the identity comparison between two G_n-torsors, hence an isomorphism before that cover by descent. On three factors both composites send $[r, s, z]$ to rsz. On any number of factors all parenthesizations are induced by this same multiplication map; its associativity gives the higher coherence as well as the three-factor identity. Raising to the ℓth power commutes with these maps. The formula for interchange has ℓ^nth power $t^m/x = y$, and sends $z\zeta$ to $(t_n^m/z)\zeta^{-1}$. Its square is the identity. Compatibility with units is the identity $(az)^{-1} = a^{-1}z^{-1}$ on the contracted products. All these are identities of actual maps of torsors. □

For a representation V of G_n, our associated sheaf convention is $(z\zeta, v) \sim (z, \zeta v)$. Choosing a local root s of a identifies $T_n(x)$ with $T_n(ax)$ by $z \mapsto sz$. Changing s to $s\zeta$ changes the associated-sheaf comparison by $\rho(\zeta)$. This is its descent datum, not an ambiguity to discard. Equivalently the comparison twists the coefficient by $T_n(a)$. In particular the two functors using x and ax are not assigned an identity natural transformation on constant representation labels. Explicitly, if $x_j = a_{ij}x_i$ and local roots s_{ij} are chosen, then on a triple overlap

$$\zeta_{ijk} = s_{jk}s_{ij}s_{ik}^{-1} \in G_n, \qquad \phi_{jk}\phi_{ij} = \rho(\zeta_{ijk})\phi_{ik}.$$

The contracted-product comparison retains precisely this transition; one cannot set $\zeta_{ijk} = 1$ merely by citing associativity. On four overlaps the corresponding cocycle identity follows from multiplication of the s_{ij}. Thus the construction descends the torsor-twisted presentations, not constant labels with their transition data erased.

Proposition 2.2 (Coefficient passage and intrinsic vertical unipotence)**.** *The torsor comparisons induce comparisons of actual associated coefficient sheaves, their tensors, duals, and evaluation maps, with the preceding descent cells. They extend to continuous adic local systems with a stable finite lattice and to their bounded derived unipotent extensions. On a strict nodal germ, the category of local coefficients with unipotent relative Kummer action is independent of an ordered node coordinate. This does not assert that its horizontal descent datum is constant on a larger overlap.*

Proof. At each finite precision a finite continuous representation of the relative ℓ-Kummer inertia factors through some G_n. The associated-sheaf construction carries the actual maps of Lemma 2.1 and their identities to maps and identities of coefficient sheaves. The same construction on $V \otimes W$ and $V^\vee$ identifies the maps with tensor and dual comparisons: the transitions on the dual are $\rho(\zeta)^{-\vee}$, so evaluation descends. There is no averaging argument here.

For a lattice $V_\bullet$, choose a factoring level separately at each precision and use a common refinement for each comparison. The torsor maps commute with all such refinements and with coefficient reduction. They therefore define morphisms of cartesian adic systems, with their already defined cells. Equivalences are tested componentwise in that category. Rationalize only afterwards. Finite Postnikov extensions are taken in the actual derived sheaf categories, where these exact tensor comparisons already exist. The finite-action derived descent input of [1] permits finite coefficient diagrams to be represented at common levels when required; a single finite level for a whole integral system is not asserted.

The relative characteristic group of $xy = t^m$ is $P_m^{\mathrm{gp}}/\mathbf{Z}[t] \simeq \mathbf{Z}$; its geometric ℓ-Kummer inertia is its hom into $\mathbf{Z}_\ell(1)$. Multiplication of a log chart by a unit does not change this group.

An ordered branch chooses a generator; interchange negates it. Unipotence means that a finite-dimensional representation admits a finite filtration with trivial relative-inertia quotients. This condition is invariant under these group identifications and under conjugation; it does not require a chosen logarithm or generator.

There is also a geometric verification that unit twists do not add vertical monodromy on a strict germ. A unit in a strictly henselian local ring with ℓ invertible has an ℓth root: lift a root in the separably closed residue field by Hensel's lemma. Applying this recursively constructs compatible roots of the unit on that strict germ. Thus its Kummer tower there is trivializable, although not canonically trivialized. On a larger overlap it need not be trivial; its transition sheaves must be retained. This argument uses a strict germ, not a claim that one finite etale cover trivializes every root level globally. □

2.2 The actual node category and branch maps

For a geometric node q, let $\mathcal{U}(T_q)$ denote the full bounded derived category on its actual geometric vanishing annulus whose cohomology local systems have a finite filtration with constant quotients. This is an intrinsic condition on actual local systems; it is not defined by being in the image of a Morita functor.

Proposition 2.3 (Node source and geometric Morita comparison)**.** *There is an equivalence induced by the actual cochain functor*

$$\mathcal{U}(T_q) \xrightarrow{R\Gamma} \mathrm{Perf}(A_q).$$

For each of the two strict punctured branches b at q, the actual nearby open-unit map

$$\beta_b : A_q \longrightarrow B_b := R\Gamma(\eta_b, Q)$$

is an equivalence of cochain algebras. The two branch orientations are opposite. These statements and their comparison functors respect unit-coordinate changes by the torsor descent above.

Proof. [1] supplies the fixed-node comparison; [2, `prop:all-thickness-nearby`] supplies the all-thickness calculation. Together they identify the actual constant-coefficient maps. The node and branch algebras have Q in degree zero, $Q(-1)$ in degree one, and no other cohomology. On a branch with parameter x the degree-one map is its primitive Kummer class. On the opposite branch y it is its negative, since $[x] + [y] = m[t]$ and the geometric base class vanishes. There is no factor m in the relative generator. These are the maps induced by the nearby open unit, not chosen cohomology isomorphisms. Consequently they are algebra equivalences.

The actual annulus cochain comparison has the same primitive Kummer class and identifies its unit endomorphism algebra with A_q. A unipotent local system is a finite extension of constant sheaves; finite Postnikov towers put every object of $\mathcal{U}(T_q)$ in the thick closure of Q. Conversely cohomology of a finite cone or retract of these objects still has unipotent monodromy. Thus the condition defining $\mathcal{U}(T_q)$ equals that thick closure. Evaluation gives the usual Morita comparison on its generator; exactness in both variables extends the mapping equivalence and essential image to the two thick closures. This proves the stated equivalence with $\mathrm{Perf}(A_q)$, on full mapping spectra.

Under $x' = ax$, the Kummer class changes by $[a]$. The torsors of a extend etale over the chart, so their specialization is the actual torsor of the specialized unit. On a strict branch this unit has compatible roots and has zero residue. This preserves the primitive branch calculation, but does not set $[a] = 0$ on a whole horizontal overlap. Lemma 2.1 and Proposition 2.2 give the associated functor comparisons and their cells before cohomology is taken. □

Lemma 2.4 (Naturality of links and pairings on overlaps)**.** *The comparisons just constructed commute with actual nearby open-unit maps, restrictions to a branch, and their Verdier mates. Their compatibility on triple overlaps is the compatibility of these actual natural transformations, including mapping homotopies.*

Proof. An etale change of chart identifies nearby cycles by etale base change. This is the comparison obtained by restricting the generic and special sheaf sites. The square involving $F \to Rj_*j^*F$ commutes because, after the open adjunction, both paths are the same open counit applied to the same coefficient map. Closed restriction gives the branch square. For a composite of restrictions this is the composite counit square. It defines the cells on strings of restrictions, so no new cells are chosen after computing stalks. The tensor comparison of the torsor functors preserves evaluation. On bounded constructible sheaves Verdier duality takes these squares to their adjoint-mate squares and preserves their composition cells; its trace is the actual trace for the geometric support map. These assertions concern sheaves already compared by actual maps, not a claim that arbitrary equivalences of cohomology lift to maps. □

3 An intrinsic supported nearby source

This section specifies what "actual coefficients" means for the first result. The source is a category of actual constructible nearby-algebra modules on Y, with a geometric supported generation statement below. It is not claimed to be a full category of generic sheaves before nearby cycles. Retaining this distinction is essential: adding arbitrary attachments changes the source, and nearby images do not in general recover arbitrary generic-fibre maps.

Definition 3.1 (Intrinsic stalk conditions)**.** Let $\mathcal{A}(Y)$ be the full subcategory of the actual stable category of H-module sheaves consisting of bounded constructible modules F such that

$$j^*F \in D^b_{\mathrm{lisse}}(U, Q), \qquad i_q^*F \in \mathrm{Perf}(A_q) \quad (q \in Z).$$

Here lisse cohomology has finite rank, with continuous geometric ℓ-adic monodromy; no global unipotence or unit-generation condition is imposed on U. All attachments are arbitrary actual H-module attachments. For a locally closed union S of this stratification use $H_S = H|_S$ and the same conditions on its retained strata. For a finer marked smooth-point stratification, its additional closed stalk algebra is Q; that is a new support profile.

This definition involves no modification, essential-image criterion, anchor, or assertion that a counit is invertible. Its node condition is expressed through an actual geometric stalk algebra. By Proposition 2.3 it has an independently identified local Kummer coefficient meaning.

Lemma 3.2 (Finite branch links with arbitrary horizontal monodromy)**.** *For $V \in D^b_{\mathrm{lisse}}(U, Q)$, the actual link*

$$L(V) = i^*Rj_*V$$

has components $L_q(V) \in \mathrm{Perf}(A_q)$. Its two branches use the actual algebra maps β_b, and $L_q(V) = \bigoplus_{b \ni q} R\Gamma(\eta_b, V)$ as an A_q-module. If V has cohomology in $[a, b]$, the link has cohomology in $[a, b+1]$.

Proof. The strict punctured neighbourhood of a node is the disjoint union of its two punctured strict branches. Each is the spectrum of the fraction field of a strictly henselian discrete valuation ring; its etale coefficients are continuous representations of its inertia group I_b. The pro-ℓ tame quotient is $\Gamma_b = \mathbf{Z}_\ell(1)$. The kernel K_b of $I_b \to \Gamma_b$ is an extension of wild pro-p

inertia by the prime-to-ℓ part of tame inertia. In residue characteristic zero the wild subgroup here is trivial. The kernel has ℓ-cohomological dimension zero. On a finite-dimensional continuous Q-representation its image is finite of order prime to ℓ: a stable lattice has an open pro-ℓ congruence subgroup, which has trivial intersection with this image. Thus its invariants are exact, with the actual finite averaging projector. This reduction is local on a branch; it does not discard the horizontal local system on U.

For a representation V_b put $W = V_b^{K_b}$. Derived invariants reduce to $C^*_{\mathrm{cts}}(\Gamma_b, W)$. Choose a generator only for this calculation. The generalized eigenvalue-one summand W_1 of its operator T is unipotent; on the complementary rational primary summand $W_{\neq 1}$, $T-1$ is invertible. The continuous cochain calculation is the two-term complex $[W \xrightarrow{T-1} W]$, so the latter summand is zero. The summand W_1 is a finite extension of trivial representations. Its cochains therefore lie in the thick closure of $C^*_{\mathrm{cts}}(\Gamma_b, Q) = B_b$, without replacing its extensions by split cohomology. They are perfect B_b-modules, and hence perfect A_q-modules under β_b. This also proves the cohomological interval. Apply finite Postnikov triangles to a bounded lisse complex. The argument uses the actual invariant and cochain maps, so identifies the module structure of the geometric link. It does not assume the horizontal monodromy itself is unipotent or that its entire cochain functor is faithful. □

Theorem 3.3 (Coordinate-independent supported source and reconstruction)**.** *For the fixed actual nodal model M, Definition 3.1 is coordinate independent and gives an equivalence with the category of homotopy-coherent triples*

$$(V, W, \alpha), \quad V \in D^b_{\mathrm{lisse}}(U, Q), \quad W \in \prod_{q \in Z} \mathrm{Perf}(A_q), \quad \alpha : W \longrightarrow L(V).$$

The equivalence retains full mapping spectra, arbitrary attachments and actual localization boundaries. It is compatible with strict etale changes of chart and with the torsor comparisons above. Moreover $\mathcal{A}(Y)$ is exactly the finite stable, idempotent-complete envelope in actual H-module sheaves generated by

$$j_! V \quad (V \text{ an actual finite-rank lisse sheaf on } U), \qquad i_{q,*} R\Gamma(T_q, E) \quad (E \in \mathcal{U}(T_q)).$$

The node modules in the second formula are transported by the actual nearby comparison to A_q-modules. Thus this envelope is geometrically specified before any model-descent theorem is used.

Proof. The nearby algebra H, its strata, and the stalk and lisse conditions are intrinsic. Etale base change identifies these data. The only chart description at a node is its local Kummer presentation, which is coordinate independent by Propositions 2.2 and 2.3. No assertion that the unit torsor is canonically trivial is needed.

Recollement holds in the ambient stable categories of module sheaves: j^*, i^* are jointly conservative; $j_!, i_*, Rj_*$ are their actual sheaf-module operations. Given a triple, form

$$F = Rj_* V \underset{i_* L(V)}{\overset{h}{\times}} i_* W.$$

Its restrictions are V and W. The branch lemma makes Rj_*V bounded constructible with perfect node stalks. Thus F satisfies the defining conditions. Conversely the open unit of a given F produces α and the canonical map to this pullback; the two restrictions show it is an equivalence. For two triples the mapping spectrum is the homotopy fibre of

$$\mathrm{Map}_U(V, V') \oplus \mathrm{Map}_{A_Z}(W, W') \longrightarrow \mathrm{Map}_{A_Z}(W, L(V')), \qquad (f, g) \longmapsto L(f)\alpha - \alpha' g.$$

Here A_Z denotes the family (A_q). In particular the homotopy in an attaching square is retained; a diagram of object values would not give this formula. The same construction gives

$$i^!F = \mathrm{fib}(W \xrightarrow{\alpha} L(V)), \qquad j_!V \longrightarrow F \longrightarrow i_*W \xrightarrow{\delta_F} j_!V[1].$$

The boundary corresponds to α under $\mathrm{Map}(i_*W, j_!V[1]) \simeq \mathrm{Map}_{A_Z}(W, L(V))$, with the boundary convention of this localization triangle.

Each displayed geometric generator satisfies the intrinsic conditions. They are preserved by finite cones and retracts. Conversely every V is obtained by finite Postnikov triangles from its lisse cohomology sheaves; every $W_q \in \mathrm{Perf}(A_q)$ is the actual cochain image of an $E \in \mathcal{U}(T_q)$ by Proposition 2.3. The localization triangle therefore places every F in the stated finite envelope. This proves both inclusions without defining that envelope by a descent or a counit condition.

On smooth strata the geometric input is the actual lisse sheaf, not just its global cohomology. At finite precision its finite etale local trivializations lift to the smooth formal neighbourhood by formal etale lifting. Applying smooth nearby specialization to these lifts recovers that sheaf. These constructions commute with coefficient reduction; they give the usual geometric local realizations of the horizontal generators. This supplies local geometric provenance, not a claim of a single global algebraic lifting for every supported object or for every morphism between them.

Finally the reconstruction is made from actual adjunction units and pullbacks. Etale restriction commutes with these operations here; Lemma 2.4 identifies the resulting chart and branch comparisons and their cells. This proves the asserted compatibility rather than merely an abstract equivalence of categories. □

Corollary 3.4 (Effective chart descent at a fixed model)**.** *The categories and reconstruction of Theorem 3.3 satisfy effective descent for finite jointly surjective strict-etale chart covers of M, with the induced finite support profile. This is descent of presentations at one model, not descent along a proper modification.*

Proof. Actual sheaf-module objects and maps descend in the stable category of modules over the sheaf algebra H; on the chart nerve the algebra is the pullback of H by etale nearby base change. This is sheaf and module descent before restricting the coefficient category. The lisse, finite-rank and constructibility conditions are etale local. At a geometric node, a chart cover induces copies of the same strict geometric germ, so perfectness of its A_q-module can be checked on that cover. Boundedness is uniform on a finite cover by taking the maximum and minimum of its finitely many bounds. Thus the ambient descent object lies in $\mathcal{A}(Y)$ whenever its chart objects do, and conversely. The descent assertion on mapping spectra is the ambient sheaf-module descent assertion, restricted to a full subcategory. The explicit reconstruction and its link functor commute with these restrictions by Lemma 2.4. The same argument applies to a finite diagram of geometric supports. It retains the overlap transition maps and does not replace them by identities; in particular it does not eliminate loop holonomy. □

Corollary 3.5 (Typed Verdier partner and a nonzero boundary)**.** *Let $D_Y = R\mathcal{H}om_Q(-, \omega_Y)$ be actual geometric Verdier duality, giving the contragredient H-action. Define $\mathcal{A}^D(Y)$ independently by the conditions*

$$N \in D^b_c(Y, H\text{-Mod}), \qquad j^*N \in D^b_{\mathrm{lisse}}(U, Q),$$
$$D_Q i_q^! N \in \mathrm{Perf}(A_q), \quad D_Q = R\,\mathrm{Hom}_Q(-, Q).$$

Then $D_Y : \mathcal{A}(Y)^{\mathrm{op}} \simeq \mathcal{A}^D(Y)$ is the actual geometric duality, with its evaluation, trace and localization mates. These statements hold on the typed support restrictions above. At any node the localization boundary for $F = H$ is nonzero.

Proof. For bounded constructible sheaves on the finite-type curve, Verdier biduality is valid and the ordinary/exceptional exchange is

$$i_q^! D_Y F \simeq D_Q i_q^* F, \qquad j^* D_Y F \simeq D_U j^* F.$$

The smooth curve duality D_U preserves bounded finite-rank lisse complexes, with its genuine dimension shift and twist. Exact finite-dimensional biduality at q proves that the two displayed stalk/costalk conditions are exchanged, in both directions. Actions are the ones obtained by transposing the actual H-action under evaluation. Thus the dual category is characterized by actual costalks, not defined as a formal opposite. Applying this same geometric duality to localization gives its mate triangle. Evaluation and trace are the ambient geometric maps; the overlap comparisons respect them by Lemma 2.4. No assertion that ω_Y is a line bundle or scalar twist is used.

For $F = H$, at a node the attaching map is $A_q \to B_{b_1} \oplus B_{b_2}$. Each component is β_b, an equivalence preserving the unit; in degree zero it is $1 \mapsto (1, 1)$. It is not zero. The mapping identification in Theorem 3.3 therefore proves that δ_H is a nonzero actual module-sheaf morphism. This test is local and does not yet construct the global loop example. □

Remark 3.6 (What this source theorem does and does not identify). Theorem 3.3 closes the local source question for the explicitly specified nearby-enhanced supported envelope. It permits arbitrary horizontal local systems and genuine attachments, without global unit generation or a coarse anchor. It does not identify this envelope with a full category of all algebraic generic-fibre coefficients; its maps are actual nearby-module maps. Such an identification would require a new source theorem and can fail if generic supports have been forgotten. It also does not identify all fine objects with pullbacks from a coarser model. The next section proves the essential-image theorem for actual curve modifications; Section 5 then proves the anchor-free profile comparison in the specified modification component.

4 Actual curve modifications and typed supported descent

An allowed map in this section is an actual proper morphism $u : M' \to M$ over the fixed trait, an isomorphism on the generic fibre and away from finitely many nodes, whose strict-etale local models are finite integer-cut modifications of $xy = t^m$. This is a geometric condition on an existing morphism; existence of such a morphism for an arbitrary proposed subdivision is not assumed. Write $u : Y' \to Y$ also for the special-fibre map, and set $F_u = u_H^* = H' \otimes^L_{u^*H} u^*(-)$ and $G_u = Ru_*$. The right adjoint uses the actual algebra map $H \to Ru_*H'$.

Definition 4.1 (The geometric fibre condition)**.** Let $C_q = u^{-1}(q)$ be the complete fibre over a modified node. Define $\mathcal{A}_u(Y')$ as the full subcategory of $\mathcal{A}(Y')$ with the following additional conditions on each C_q. On every open exceptional component $\mathbf{G}_m$, the cohomology local systems are unipotent. At every incidence within C_q, the individual module attachment to its punctured branch link is invertible, using the actual branch algebra map. There is no invertibility requirement on attachments from C_q to the exterior open strata of Y'. These conditions are imposed on the restriction to the complete closed fibre, not on its costalk. For a support $S \subseteq Y$ use $S' = u^{-1}S$ and the same conditions on fibres contained in S'.

The fibre is a chain, including its terminal nodes. This definition uses its genuine strata, unipotent monodromy and branch maps; it is not a definition as the image of F_u. No condition is imposed on the global dual graph of Y or on horizontal monodromy outside the exceptional chains.

Lemma 4.2 (Actual algebra unit and the chain coefficient category)**.** *The actual unit $H \to Ru_*H'$ is an equivalence of nearby algebras. For every modified node q,*

$$F_q : \mathrm{Perf}(A_q) \longrightarrow \mathcal{C}(C_q), \qquad W \longmapsto H_{C_q} \otimes^L_{A_q} W$$

is an equivalence onto the category specified by the fibre conditions in Definition 4.1. Its inverse is $R\Gamma(C_q, -)$, with the A_q-action given by the actual algebra unit.

Proof. Proper nearby base change identifies the unit with the map below. At torsion level it follows from ordinary proper base change [6, Expose XII, Theorem 5.1], applied to the proper scheme map u and the special-fibre immersion, together with composition of generic direct images in the definition of geometric nearby cycles. We use its constructible adic comparison and rationalization [7, 8] for these finite-type trait models with ℓ invertible. Thus the unit is $R\Psi_M Q \to Ru_* R\Psi_{M'} u_\eta^* Q$. The generic map is an isomorphism, so this is an equivalence. The comparison is the mate of that generic pullback identity; its multiplicative structure is therefore that of the actual nearby algebra map. Proper base change restricts it to $A_q \simeq R\Gamma(C_q, H_{C_q})$.

For $W = A_q$, the coefficient unit $W \to R\Gamma(C_q, F_q W)$ is just this map. The projection map extends the assertion to finite cones and retracts, hence to every perfect W. This proves full faithfulness on perfect modules. The local chain reconstruction of [2, `thm:cell-reconstruction`] identifies the independently specified unipotent, branchwise-invertible fibre category with $\mathrm{thick}(H_{C_q})$. Its hypotheses hold on each actual integer-cut fibre; its argument concerns that chain, not the global curve. The image of the fully faithful F_q is thick and contains H_{C_q}, hence is exactly this category. The torsor and branch comparisons of Section 2 identify the conditions under changes of these local charts. □

Theorem 4.3 (Pairwise curve descent with arbitrary exterior attachments)**.** *For every actual modification just specified and every locally closed coarse support S that is a union of the strata of Definition 3.1, with its complete inverse image S', there is an actual adjoint equivalence*

$$F_u : \mathcal{A}(S) \xrightarrow{\sim} \mathcal{A}_u(S'), \qquad G_u = Ru_{S,*}.$$

Its unit is the actual module-adjunction unit. It preserves all mapping spectra, arbitrary exterior attachments and their localization boundaries. No global unit-generation hypothesis or forest condition on the whole curve is required.

Proof. It suffices to give the argument on Y; restriction to a complete inverse-image support uses the identical proper base-change squares. Let U be the complement of the coarse nodes, Z their union, $Z' = u^{-1}Z$, and write j, i, j', i' for the immersions. Unmodified nodes have point fibres and are included with identity transitions. Over U the map is the identity. Put

$$L(V) = i^* R j_* V, \qquad L'(V) = i'^* R j'_* V, \qquad F_Z = \prod_q F_q, \qquad G_Z = Ru_{Z,*}.$$

Proper base change and composition of right adjoints give the actual comparison $\lambda_V : G_Z L'(V) \xrightarrow{\sim} L(V)$. It applies to arbitrary lisse V on the exterior U. The link $L'(V)$ is an ambient module object; it need not satisfy the fibre conditions and is not replaced by a cartesian object.

Define $\chi_V : F_u R j_* V \to R j'_* V$ to be adjoint to the identity of V on U. Its closed restriction is $\kappa_V : F_Z L(V) \to L'(V)$. The adjoint of κ_V under the ambient adjunction $F_Z \dashv G_Z$ is λ_V^{-1}: indeed the composite

$$R j_* V \longrightarrow G_u F_u R j_* V \xrightarrow{G_u \chi_V} G_u R j'_* V \simeq R j_* V$$

restricts to the identity on U. Full faithfulness of $R j_*$ identifies it with the identity, with its adjunction homotopy. Proper base change carries this unit identity to the closed square. Thus, with ϵ_Z the actual ambient counit,

$$\kappa_V = \epsilon_{Z,L'(V)} F_Z(\lambda_V^{-1}).$$

The actual pullback of a coarse triple (V, W, α) therefore has triple

$$(V, F_Z W,\ \epsilon_{Z,L'(V)} F_Z(\lambda_V^{-1}\alpha)).$$

This formula follows from the open unit defining the attaching map; it is not a freely chosen gluing equivalence.

Lemma 4.2 identifies the possible closed objects with the fibre category. The ambient adjunction identifies the attaching mapping spectra by

$$\mathrm{Map}(F_Z W, L'(V)) \simeq \mathrm{Map}(W, G_Z L'(V)) \simeq \mathrm{Map}(W, L(V)).$$

These identifications preserve the difference map in the mapping-fibre formula of Theorem 3.3; hence they identify the full mapping spectra of triples, including their extension homotopies. They also give an inverse on all fine triples satisfying the geometric fibre condition. This proves the asserted equivalence with actual F_u and G_u.

Alternatively its unit is checked on the jointly conservative coarse strata: the identity over U and the perfect-module unit of Lemma 4.2 at each q. Bounded constructibility of the selected pullback and pushforward follows from the finite chain calculation and the same recollement: all exterior objects and links are bounded constructible, with finitely many fibres. This argument does not require a global generator and leaves the entire exterior local system, including its monodromy, unchanged. Finally $F_u j_! V \simeq j'_! V$ and $F_u i_* W \simeq i'_* F_Z W$. The mapping comparison identifies the actual connecting morphism between these objects. It therefore compares localization triangles, not only their three object values. □

Proposition 4.4 (The operation diagram has two coefficient types)**.** *Let $\mathcal{A}_u^D(S')$ be the bounded constructible modules N whose actual $D_{S'}N$ satisfies the intrinsic fibre conditions above. Equivalently the fibre test is applied to $D_{C_q} i^!_{C_q} N$; this uses actual costalks. The supported transition*

$$F_u^D = D_{S'} F_u^{\mathrm{op}} D_S : \mathcal{A}^D(S) \xrightarrow{\sim} \mathcal{A}_u^D(S')$$

has inverse actual proper direct image. Ordinary pullback compares support restrictions, extension by zero and closed pushforward on the ordinary types. The dual transition compares their Verdier mates on the supported types. The pairings and trace comparisons are the actual geometric ones.

Proof. The costalk characterization follows from $i^*_{C_q} D_{S'} N \simeq D_{C_q} i^!_{C_q} N$. Proper Verdier duality gives $D_S G_u \simeq G_u D_{S'}$, with the action through the same nearby-algebra map. Conjugating the proved adjunction equivalence therefore gives the stated supported equivalence and its inverse, identified by this actual proper-duality map.

Restriction and extension-by-zero squares for F_u are actual module pullback squares on complete inverse-image supports. The closed-pushforward square is the corresponding closed

support base-change square. Applying actual Verdier duality gives the Rj_* and exceptional-restriction squares with transition F_u^D, with all types changed accordingly. Naturality of evaluation and proper trace identifies their pairing squares; the identities of adjunction and biduality identify the localization mates. No identification $F_u = F_u^D$ in the fine ambient category is asserted. □

Example 4.5 (Ordinary pullback does not preserve the raw right open image). Take the thickness-two node and its one-cut modification, with exceptional fibre C, and let $j : U = Y \setminus \{q\} \hookrightarrow Y$. For $V = Q_U$, its coarse link is $A_q \oplus A_q$. Thus on the open exceptional component

$$(F_u R j_* Q_U)|_{C^\circ} = Q_{C^\circ}^{\oplus 2}.$$

But $Rj'_* Q_U$ restricts to zero on C°: this open subset of Y' is disjoint from U. Hence the actual comparison $F_u R j_* Q_U \to Rj'_* Q_U$ is not an equivalence. The latter link must remain in the ambient category in the preceding proof. The supported-type comparison of Proposition 4.4 is unaffected. An untyped assertion that ordinary pullback intertwines every raw six-operation term is false.

Proposition 4.6 (Coherence for strings of actual modifications)**.** *For a composable string of allowed actual modifications, the comparisons above have the composition cells of the actual module pullback/direct-image adjunctions, with their support and Verdier mates. The image for a composite is the iterated image; it is also characterized directly by the fibre condition for the composite.*

Proof. The algebra maps $u^* H \to H'$ come from the same generic pullback identities and nearby adjunctions. Thus on strings they form the actual coherent diagram of algebra maps. Extension of scalars and derived tensor associativity give $F_{uv} \simeq F_v F_u$, with the associativity cells of a single iterated tensor product. For the external link comparison both χ_{uv} and $\chi_v F_v \chi_u$ are adjoint to the same open identity. Their specified adjunction identification gives its composition homotopy and, on longer strings, its higher cells. The same applies to λ by right-adjoint composition. The mapping-fibre proof consequently transports each attachment with these cells. Applying the pairwise theorem to the composite identifies its direct geometric fibre condition with the iterated image. Verdier conjugation supplies the supported cells. This proves coherence for actual strings without transporting a preselected anchor's entire operation diagram. □

5 Geometric profiles and anchor-free descent

We now fix a connected component $\mathscr{M}$ of the category of actual nodal models and integer-cut modifications considered above. All models have the specified identification of their generic curve, and all morphisms induce its identity. Connected means connected by a finite zigzag of these actual maps. It does not mean that all models of a given generic curve are admitted. We order by refinement, so an arrow in the indexing category $M \to M'$ denotes a geometric map $M' \to M$.

Lemma 5.1 (A canonical finest model in the integer-cut component)**.** *The component $\mathscr{M}$ has a terminal refinement $\widetilde{M}$, unique up to the unique isomorphism inducing the generic identity. Its node thicknesses are all one. It can be constructed by repeatedly blowing up the reduced nonregular locus of the total space of any model in $\mathscr{M}$. Every actual integer-cut modification has the same terminal refinement.*

Proof. At a node $xy = t^m$ the total space is regular exactly when $m = 1$. For $m \geq 2$ blow up the ideal (x, y, t). On the x-chart put $t = xw$, $y = xv$. The strict-transform equation

is $v = x^{m-2}w^m$, so this chart is $\operatorname{Spec} R[x,w]/(xw-t)$ with v eliminated. The y-chart is identical. On the t-chart put $x = tu$, $y = tv$; its equation is $uv = t^{m-2}$. For $m = 2$ this last chart is $uv = 1$; for $m = 3$ it has thickness one. These equations follow in the blow-up charts by removing the torsion in the Rees presentation; the original node is integral, so they are the charts of the blow-up itself. Thus for $m > 2$ the outer cuts are $1, m-1$ and the remaining middle thickness is $m-2$; for $m = 2$ there is the single cut 1. Iteration gives exactly all integer cuts $1, \dots, m-1$ and terminates.

The nonregular locus is a finite reduced closed subscheme, and its blow-up is an actual globally defined projective modification. The chart computation is etale local and is unchanged by multiplication of coordinates by units or by interchange of branches. It therefore constructs the full resolution globally, without gluing hypothetical modifications or invoking a contraction theorem.

For a partial cut set, resolving each remaining gap inserts precisely the missing integer cuts. The full-cut comparison in [2, `prop:cut-models`] hence identifies its resolution with that of the original node, compatibly with the map to the original model. These local identifications glue: two maps between the separated models agreeing on the dense generic fibre agree everywhere. The models here are reduced and flat, and the generic fibre is schematically dense. The same uniqueness removes any dependence on a zigzag connecting models in $\mathscr{M}$. Consequently the resolution of any one member resolves every member and gives its unique refinement arrow to the resulting $\widetilde{M}$. Resolving again does nothing, so this is terminal. □

Write $\widetilde{Y} = \widetilde{M}_s$ and let $\widetilde{G}$ be its finite dual graph, retaining loop edges and the genera of normalized components. Each component defines its normalized divisorial valuation on the function field of the fixed generic curve. These valuations identify components across all presentations.

Definition 5.2 (A marked geometric constructibility profile)**.** A profile $\mathfrak{p}$ in $\mathscr{M}$ consists of the following geometric data, chosen before any coefficient category. Choose a subset P of these divisorial valuations. Every unmarked component of $\widetilde{Y}$ must have normalization $\mathbf{P}^1_k$ and exactly two incident branches. Require each connected component of the subgraph induced by unmarked vertices to be a path, not a cycle. In particular an unmarked isolated self-loop is excluded. All vertices of positive genus or of valence different from two must therefore be marked.

The profile pieces on $\widetilde{Y}$ are the smooth open strata U_v of marked components, and the closed blocks C_e consisting of a maximal unmarked path together with all its incident nodes. A node directly between marked components is itself a block. A path may return to the same marked component; it still has two terminal branches and is allowed. These pieces partition $\widetilde{Y}$.

Choose finitely many closed subsets that are unions of whole profile pieces, include $\varnothing$ and $\widetilde{Y}$, and take their finite closure under unions and intersections. The support diagram contains the differences of these closed subsets, with the induced locally closed inclusions and open/closed decompositions. This is still a finite diagram: the finitely many profile pieces give only finitely many subsets. In particular the diagram contains support intersections used in kernel composition. A presentation is a model $M \in \mathscr{M}$ in which every marked valuation remains a component.

This definition uses a canonically constructed fine model, not a chosen coarse model. Equivalently its markings are specified geometric valuations and residual branches. Their coefficients and descent properties are not used to choose them. Every profile has a presentation, namely $\widetilde{M}$ itself. We do not assume that contracting all its unmarked paths produces another model.

For a presentation let $r_M : \widetilde{M} \to M$ be the canonical map. Its exceptional fibres contain no marked component and lie inside single blocks. Let $U_{v,M}$ and $C_{e,M}$ be the images of the profile pieces. The former are identified with U_v, and the latter are closed chains, possibly just a node. The full inverse image of $C_{e,M}$ is C_e. The specified supports are likewise complete inverse images: the marked closed subsets are saturated for r_M, so properness takes them to closed subsets and their differences to the required locally closed supports. This proves existence of all these geometric support presentations.

Definition 5.3 (Profile coefficients)**.** Let $\mathcal{A}_{\mathfrak{p}}(M,S)$ be the full subcategory of $\mathcal{A}(S_M)$ whose restriction to every block $C_{e,M}$ retained in the support has unipotent cohomology on its open rational components and invertible individual internal branch attachments. At a block consisting of a point this is the existing perfect-stalk condition. The marked open strata retain arbitrary horizontal lisse coefficients, and attachments between a block and marked open strata are arbitrary. Define the supported type by actual Verdier costalk conditions: N belongs to $\mathcal{A}^D_{\mathfrak{p}}(M,S)$ when its actual $D_{S_M}N$ satisfies these geometric ordinary-type conditions. On a closed block the test is on $D_{C_{e,M}} i_e^! N$.

Lemma 5.4 (Support operations preserve the geometric profile)**.** *Let $e : S \hookrightarrow T$ be an inclusion in the finite support diagram. The actual operations e^* and $e_!$ preserve the ordinary profile categories. In particular closed pushforward does so, and an open/closed decomposition gives its actual localization triangle inside these categories. Their Verdier mates $e^!$ and Re_* preserve the supported profile categories. These operations commute with the corresponding typed modification comparisons on complete inverse-image supports.*

Proof. Every support is a union of whole profile pieces. Thus its intersection with a marked open stratum is either that entire stratum or empty; its intersection with a closed block is either that entire block or empty. Restriction preserves the conditions on each retained piece. For extension by zero, the ordinary restriction to a piece outside S is zero, by base change for an immersion; on a piece in S it is the original restriction. Consequently an unmarked block has either its original unipotent, branchwise-invertible coefficient or the zero coefficient, and marked open strata have lisse restrictions. Node stalks remain perfect. Attachments at the boundary of S are exterior attachments between pieces, on which the definition imposes no invertibility requirement. No internal branch of a block is cut. The underlying operations are bounded constructible on this finite stratification. This proves preservation, including zero blocks.

For a closed i and complementary open j, the actual triangle $j_! j^* F \to F \to i_* i^* F$ has all its terms in the ordinary profile category by the preceding argument; its connecting map is unchanged. Actual geometric Verdier exchange gives the supported operations and the mate triangle. Finally Proposition 4.4 compares these ambient operations by actual maps on full inverse-image supports. The preservation just proved allows those same maps to restrict to the profile categories. This last step cannot be replaced by preservation of the larger ambient category alone. □

Lemma 5.5 (The same profile selects the same fine objects)**.** *For every presentation M, actual pullback to $\widetilde{M}$ induces equivalences*

$$F_{r_M} : \mathcal{A}_{\mathfrak{p}}(M,S) \xrightarrow{\sim} \mathcal{A}_{\mathfrak{p}}(\widetilde{M},S),$$

and similarly for the supported types using $F^D_{r_M}$. These are statements about independently specified geometric conditions on both sides.

Proof. Each block is a chain with two terminal branches, not a circular fibre. Its unipotent, branchwise-invertible module category is thick($H_{C_{e,M}}$). To see that the chain result

used earlier applies without assuming a contracted coarse model, use its actual recollement presentation: on each open rational component, continuous Kummer cochains identify the unipotent category with perfect modules over its unit cochains; each puncture restriction is the primitive Kummer equivalence. At a node use Proposition 2.3. Requiring each branch map to be invertible therefore joins these perfect-module categories by equivalences along a finite incidence tree. Successively removing terminal vertices identifies the resulting category with any one of these perfect-module categories. The actual unit restricts to the unit at every vertex, hence generates the result. This is an argument on a single closed block, never on the possibly cyclic global graph. Actual torsor twists are retained in its equivalences.

The pairwise theorem applies to r_M. On a block its pullback is fully faithful and carries $H_{C_{e,M}}$ to H_{C_e} by actual extension of scalars. Thus it identifies their two thick unit categories. Every fine profile object satisfies the fibre condition for r_M, because each fibre is a subchain of one block. The pairwise theorem first descends this object to M; the just proved block comparison shows that its descended restriction has the required block condition. Conversely a coarse profile object pulls back to the fine block category. Outside the blocks the map is the identity. The external-attachment mapping equivalence of Theorem 4.3 retains all maps joining these blocks to the marked open strata. This proves both image inclusions and full faithfulness, not just equality of object restrictions. Actual proper duality proves the supported assertion. □

Theorem 5.6 (A: anchor-free typed supported Morita descent)**.** *Fix an integer-cut component $\mathscr{M}$ and a marked geometric profile $\mathfrak{p}$ as above. The systems*

$$\{\mathcal{A}_{\mathfrak{p}}(M,S), \mathcal{A}^{D}_{\mathfrak{p}}(M,S)\}_{S}$$

for all presentations admit canonical coherent comparisons independent of a coarse anchor and of the choice of common refinement. The comparisons retain full mapping spectra, actual localization boundaries, arbitrary horizontal and loop monodromy, and the typed geometric Verdier pairings and trace maps. The ordinary support operations and their supported Verdier mates are those of Proposition 4.4; no untyped six-operation claim is made. These comparisons in particular give a multi-object Morita-invariant supported system, without a global unit generator.

Proof. The terminal refinement of Lemma 5.1 is a presentation of every profile. It is therefore a common refinement of every finite family of presentations; the same holds for any finite family of common refinements. Lemma 5.5 proves, by the geometric block condition, that the images of all presentations in it are exactly the same category. This is the essential step that equality of support labels alone did not prove.

For presentations M, N put $F_M = F_{r_M}$ and $G_M = Rr_{M,*}$ on the indicated ordinary types. The comparison is $T_{N\leftarrow M} = G_N F_M$. Its composite with $T_{L\leftarrow N}$ is compared to $T_{L\leftarrow M}$ by the actual counit $F_N G_N \to \mathrm{id}$ on $\mathcal{A}_{\mathfrak{p}}(\widetilde{M}, S)$. The counit is invertible on that geometrically identified category. The units and triangle identities give the identity comparisons; composition of these same adjunctions on strings gives associativity and all higher cells, as in Proposition 4.6. Thus the result is not obtained by choosing unrelated equivalences.

If another actual common refinement L is used, it refines further to $\widetilde{M}$. The string comparisons and the same counit identify the comparison through L with the displayed one. For any finite diagram of such choices these identifications are the restrictions of the single coherent adjunction diagram through the terminal refinement. This proves independence of common refinement with its higher comparisons. The uniqueness of the generic-identity isomorphism of terminal refinements makes the construction independent of the model used to construct $\widetilde{M}$.

For each specified support, r_M uses its complete inverse image. Lemma 5.4 proves preservation of the profile categories by the support operations themselves. Actual support squares and their localization connecting maps were compared in the pairwise theorem. Applying the argument simultaneously to the finite support diagram gives its ordinary operation comparisons. Actual Verdier conjugation and proper trace give the supported comparisons and mixed pairing cells, with their types retained. An equivalence of these small idempotent-complete stable categories is a Morita equivalence in the multi-object sense. Nothing replaces a marked stratum's full coefficient category by its cohomology algebra, and no step removes the gluing around a cycle. This proves all the assertions in the stated domain. □

Remark 5.7 (The precise model-independence obtained). Theorem 5.6 settles the coefficient-and-support presentation problem for these geometric profiles inside a connected component of actual integer-cut modifications. It neither claims that all models of one generic curve belong to that component nor allows arbitrary rational cuts over a fixed trait. The canonical fine model is available because this restricted modification class has a finest refinement. In a larger model category that fact requires a new theorem and may fail. Changing the marked valuation set changes the constructibility profile; new fine objects are not silently admitted into an old profile. The source remains the actual nearby-enhanced supported envelope, not all generic sheaves with all their original maps.

6 Geometric kernels with horizontal coefficients

Fix the component and profile of Theorem 5.6. Close its finite support list under the intersections used below; these are again differences of closed saturated unions. Products of supports are over k. Write $\mathcal{C}_a = \mathcal{A}_{\mathfrak{p}}(M,S)$ for $a = (M,S)$, with actual algebra H_a. The public enhanced six-operation and adic formalism is that of [7, 8], used here on finite-type scheme supports with ℓ invertible and geometric rational coefficients. The ambient extension $\mathscr{D}(S) = \mathrm{Ind}(D^b_c(S,\mathbf{Q}_\ell))$, its module operations and its comparison with actual bounded module objects are the constructions of [3], `lem:bounded-ambient` and `prop:ambient-kernel-types`. Neither input asserts boundedness of arbitrary relative tensors over a nearby algebra. Profile preservation and boundedness of the specific convolution integrands are proved below; all generated expressions are proved bounded before Verdier duality is applied.

Definition 6.1 (An independently specified geometric kernel source)**.** For $a = (M,S)$ and $b = (N,T)$ with N refining M, kernels are actual $(p_S^* H_a, p_T^* H_b)$-bimodule sheaves on $S_M \times_k T_N$. Our generators are:

1. the diagonal $\Delta_* H_a$;
2. graphs and transposed graphs of the specified support inclusions within one model, with their regular coefficients, and the actual diagonal localization triangle and its arrows;
3. for each actual refinement $u : N \to M$, the transposed proper graph U_u with coefficient H_N, acting by F_u;
4. within each model, every rectangle $C \boxtimes_Q E$, where $C \in \mathcal{C}_a$, $E \in \mathcal{C}_b$ are independently defined profile coefficients on the two supports.

Take finite composable convolutions, finite cones, shifts and retracts, and retain all actual ambient kernel mapping spectra between the resulting objects. There is no restriction by equality of actions, preservation of coefficients or invertibility of a counit. Before localization,

model changes in a word are all refinements. Reverse proper graphs will be used for comparison, not declared to be generators or inverse kernels in this ambient source. In particular no boundedness of the bare projection kernel $U_u \star V_u$ is assumed. The composites used below are evaluated with their specified coefficients and proved bounded by the controlling formulas; only those bounded composites are subjected to Verdier duality.

The action and convolution are the geometric operations

$$\Phi_P(M) = Rp_{T,!}(p_S^* M \otimes^L_{p_S^* H_a} P), \qquad L \star P = Rp_{13,!}(p_{12}^* P \otimes^L_{p_2^* H_b} p_{23}^* L).$$

The regular diagonal is their unit. Tensor products and maps here are derived module operations, not tensors of cohomology groups.

Lemma 6.2 (Coefficient tensors and the required projection map)**.** *For $C, D \in \mathcal{C}_a$, the complex $C \otimes^L_{H_a} D$ is bounded constructible and belongs to $\mathcal{C}_a$. Its compactly supported cohomology is perfect over Q. For an actual refinement $u : N \to M$, profile coefficients C on M and D on N satisfy the actual projection equivalence*

$$C \otimes^L_{H_M} G_u D \xrightarrow{\sim} G_u(F_u C \otimes^L_{H_N} D).$$

These statements also hold on the specified complete support restrictions.

Proof. On a smooth stratum $H = Q$, and the complexes are lisse of bounded finite rank. At a node both stalks are perfect over A_q. Their relative tensor is therefore bounded with finite-dimensional cohomology. There are finitely many strata and a uniform bound on each lisse stratum, so this is a bounded constructible tensor on the whole support. On an unmarked block, the actual chain comparison with perfect modules is monoidal: restriction, regular module tensor, and the branch equivalences are the actual tensor maps. Thus the block condition is preserved. This proves membership in the profile category. Finiteness of compactly supported cohomology for a finite-type curve and bounded constructible rational coefficients then gives a perfect Q-complex.

Check the displayed projection map on geometric stalks using proper base change. Off exceptional fibres it is the identity. At a contracted node put $E_q = u^{-1}(q)$ and $C_q = i_q^* C$. The map is

$$C_q \otimes^L_{A_q} R\Gamma(E_q, D|_{E_q}) \longrightarrow R\Gamma(E_q, H_{E_q} \otimes^L_{A_q} C_q \otimes^L_{H_{E_q}} D|_{E_q}).$$

The coefficient stalk C_q is perfect over A_q. The map is an equivalence on A_q, then on finite cones and retracts. This proves it on all stalks, with the actual projection map and its coefficient actions. We have not assumed that an arbitrarily attached sheaf is globally tensor-dualizable. □

Proposition 6.3 (Control of generated kernels and their actions)**.** *Every generated kernel is bounded constructible; its action preserves the profile category and has bounded constructible integrand. For every composable pair of generated kernels, the relative tensor integrand defining their convolution is also bounded constructible. Geometric convolution is defined on this source with its usual associativity and unit cells.*

Proof. Here is a controlling family of actual kernels. For $M \preceq N$ and a support $W \subseteq S \cap T$, let

$$G^{S,T}_{M,N}(W) = \gamma_{W,!} H_{W_N}, \qquad \gamma_W : W_N \longrightarrow S_M \times T_N, \quad w \longmapsto (u(w), w).$$

Also use rectangles $R_{M,N}(C, E) = C \boxtimes_Q E$ with $C \in \mathcal{C}_{(M,S)}$, $E \in \mathcal{C}_{(N,T)}$. This controlling family may include objects beyond the initial rectangle list; it is specified geometrically, not by their actions. All generators lie in its finite stable envelope. The support intersections

are in the finite diagram by construction, and Lemma 5.4 supplies the required preservation by support restriction and extension by zero.

We first check relative tensor integrands, before applying direct image. Pull the two kernels to the threefold product. For two graphs their supports meet along the composite graph, and the regular intermediate algebra cancels in the relative tensor. The resulting coefficient is the final regular algebra, hence bounded. For a graph followed by a rectangle the same cancellation leaves the restriction of its first coefficient to the intermediate graph and the external second coefficient. For a rectangle followed by a graph it leaves the external first coefficient and the extension-of-scalars pullback of the middle coefficient; this pullback is a bounded selected coefficient by Theorem A. For two rectangles the middle factor is $E \otimes^{L}_{H_{T_N}} C'$, bounded constructible by Lemma 6.2. All remaining factors are external bounded constructible Q-complexes. Extensions by zero along the graph support immersions preserve this boundedness. These identifications are the actual support tensor maps: on stalks, factors outside the intersecting supports are zero, and on the intersection they are precisely the regular cancellations just described. Stalks detect equivalences and commute with relative tensor. Each coefficient family has finitely many strata, so the bounds are uniform for each such integrand. This does not assert a common bound for all objects of the category or finite Tor-amplitude for every ambient bimodule.

Graph–graph convolution is the graph on the support intersection, with the composite actual refinement. Regular factors cancel in the relative tensor. This is the actual support base-change map; the support tensor identity can be checked on scalar stalks, as in [3, `lem:support-intersection`]. For the other three cases the formulas, with well-typed supports, are as follows:

$$\begin{aligned} R_{N,L}(C,E) \star G^{S,T}_{M,N}(W) &\simeq R_{M,L}(e_{W_M,S_M,!}G_{u_W}(C|_{W_N}),E),\\ G^{T,V}_{N,L}(Z) \star R_{M,N}(C,E) &\simeq R_{M,L}(C, e_{Z_L,V_L,!}F_{u_Z}(E|_{Z_N})),\\ R_{N,L}(C',E') \star R_{M,N}(C,E) &\simeq R\Gamma_c(T_N, E \otimes^{L}_{H_{T_N}} C') \otimes_Q R_{M,L}(C,E'). \end{aligned}$$

The first is proper graph base change followed by direct image of the first coefficient; the second is graph pullback and regular cancellation on the target coefficient. The third is compact-support base change and projection along the middle factor. Thus these are kernel equivalences with both outer actions, not conclusions inferred from actions. Lemma 6.2 and Theorem A show that every coefficient on their right sides remains in the profile category and that the scalar complex in the last line is perfect. Support restriction and extension by zero are the proved ordinary typed operations. The same scalar projection maps, or their relative bar resolutions, show naturality with both module actions.

The graph action is $e_{W_N,T_N,!}F_{u_W}e^{*}_{W_M,S_M}$, and a rectangle acts by

$$\Phi_{R_{M,N}(C,E)}(M_0) = R\Gamma_c(S_M, M_0 \otimes^{L}_{H_{S_M}} C) \otimes_Q E.$$

These preserve the profile source by the same lemma. Their integrands are respectively the actual graph pushforward of a bounded selected pullback and the external product $(M_0 \otimes^{L}_{H_{S_M}} C) \boxtimes_Q E$, hence bounded constructible. The three convolution formulas and the graph formula put every finite word in the finite stable envelope of the controlling family. For two fixed objects in this envelope the integrand is obtained by finite cones and retracts from the pairwise integrands checked above: relative tensor is exact in each variable. Thus boundedness holds before direct image for all generated pairs, not merely for the values of their convolution or their actions. Scalar finite-type shriek direct image then has bounded constructible values here. The same exactness extends the action and all comparisons to finite cones and retracts, retaining all actual ambient maps. Fubini and tensor associativity

are the actual maps before restriction to this source, so supply its coherent composition. No boundedness assertion on arbitrary ambient bimodules is needed. □

6.1 Actual kernel descent for one model change

For one actual modification $u : M' \to M$, use its restrictions on both endpoint supports. Write U_a, U_b for the pullback graphs and V_a, V_b for their reverse proper graphs, acting by G_u. For a generated kernel P within M' define

$$T_u(P) = V_b \star P \star U_a \simeq R(u_S \times u_T)_* P.$$

The algebra actions on the last expression are restricted through the actual nearby algebra maps. This is simultaneous coarsening of a kernel on one model; arbitrary independent contractions of its two endpoints are not part of this assertion.

Relation to the anchored kernel theorem. The one-sided counit mechanism below, and the insertion of that counit to obtain strong convolution, already appear in [3], `thm: one-sided-descent`. That result treats a fixed local anchored four-model system and its restricted coefficient rectangles. Here the source is selected by geometric profiles, with arbitrary horizontal lisse coefficients on the marked strata. The preceding bounded-link, relative-tensor and convolution-integrand arguments verify the enlarged domain; the proof below verifies the actual counit on its generators. The further increments are independent endpoint changes on the rectangular source (Theorem 7.1) and finite etale spans in Cartesian presentations (Theorem 7.5), with a separate span-counit calculation. Neither follows merely by citing the anchored theorem, and neither asserts descent for arbitrary coupled kernels.

Theorem 6.4 (B: one-sided kernel descent and strong convolution)**.** *For the geometric generated source within M', T_u takes values in the source within M. The actual target counit induces an equivalence of kernels*

$$\alpha_P : U_b \star T_u(P) = U_b \star V_b \star P \star U_a \xrightarrow{\epsilon_b \star 1} P \star U_a.$$

Consequently T_u has invertible unit and composition maps

$$\Delta_* H_{S_M} \xrightarrow{\sim} T_u(\Delta_* H_{S_{M'}}), \qquad T_u(L) \star T_u(P) \xrightarrow{\sim} T_u(L \star P).$$

These statements are natural on full actual kernel mapping spectra and compatible with localization and strings of actual modifications.

Proof. For the controlling graph $G_{M',M'}(W)$, proper product direct image is $G_{M,M}(W)$: its coefficient comparison is exactly $H_{W_M} \xrightarrow{\sim} Ru_{W,*} H_{W_{M'}}$. For a rectangle it is

$$T_u(C' \boxtimes E') \simeq G_u C' \boxtimes G_u E'$$

by proper product base change and the scalar projection formula. Theorem A puts both resulting coefficients in their coarse profile categories. These comparisons imply preservation of the generators. Preservation of their convolutions will also follow from the counit calculation below, so is not being assumed at this step.

For a graph, both sides of α_P identify with the same graph on $W_{M'}$ over $S_M \times T_{M'}$. On that graph α_P is the coefficient counit $F_{u_W} G_{u_W} H_{W_{M'}} \to H_{W_{M'}}$. The actual algebra unit and the adjunction triangle make this the inverse pulled-back unit, hence an equivalence.

For a rectangle $P = C' \boxtimes E'$, the actual graph projection and base-change comparisons give the vertical equivalences in

$$\begin{array}{ccc} U_b \star T_u(P) & \xrightarrow{\alpha_P} & P \star U_a \\ \sim\downarrow & & \sim\downarrow \\ G_u C' \boxtimes F_u G_u E' & \xrightarrow{1\boxtimes \epsilon_{E'}} & G_u C' \boxtimes E'. \end{array}$$

The square commutes by naturality of these projection and base-change comparisons with respect to the target adjunction counit, using the same enhanced adjunction and convolution associators as in the definition of α_P. The right vertical comparison is proper graph projection on the source factor, using the coefficient projection map of Lemma 6.2; it is a kernel calculation on the product, not only a calculation of its global action. Under these specified comparisons the map α_P is $1 \boxtimes \epsilon_{E'}$. Theorem A identifies E' with the pullback of its actual proper direct image, by the independently proved block condition, so this counit is invertible. This proves both the map and its invertibility on the controlling family.

Every finite convolution in the one-model source is in its finite stable controlling envelope by Proposition 6.3. The two sides of α are exact functors on actual kernel categories and the comparison is their natural counit transformation. Its equivalence locus is therefore closed under finite cones and retracts; the calculation proves it for every generated kernel. Insert this same middle counit to obtain

$$V_c \star L \star U_b \star V_b \star P \star U_a \longrightarrow V_c \star L \star P \star U_a.$$

It is an equivalence by α_P and is precisely the strong composition comparison for T_u. This now proves convolution preservation in the generated source. The unit comparison is the actual algebra unit on the diagonal, already calculated above. The triangle identities and composition of the same counits give its unital and associative cells, including higher cells. Proper pushforward and its support base-change mates identify these maps under composed modifications. Since all are natural transformations on actual kernel categories, this includes their maps and localization connecting morphisms, not merely their isomorphism classes. □

6.2 Morita realization and actual Verdier mates

Theorem 6.5 (Model-independent realization of the generated source)**.** *The global source of Definition 6.1 has an enhanced multi-object Morita realization preserving units and convolution, with actual geometric Verdier mates. It sends each refinement graph to an equivalence and therefore factors through localization at these graphs. After the canonical identifications of Theorem A, its coefficient target and its simultaneous-coarsening comparisons are independent of model presentation and of common refinement. No equivalence or full faithfulness of the whole kernel categories is asserted.*

Proof. For a kernel $P : a \to b$ its Morita bimodule, without choosing any single generator, is the enriched Yoneda module

$$\mathbb{M}_P(X, Y) = \mathrm{Map}_{\mathcal{C}_b}(Y, \Phi_P(X)), \qquad X \in \mathcal{C}_a,\ Y \in \mathcal{C}_b.$$

Each target module is representable and hence perfect in the multi-object Morita sense. This follows from the proved preservation of $\mathcal{C}_b$, not from a stipulated perfectness condition on kernels. The actual geometric action on all kernel mapping spectra makes this an enhanced functor. Enriched Yoneda evaluation and geometric Fubini give $\mathbb{M}_P \otimes_{\mathcal{C}_b} \mathbb{M}_L \simeq \mathbb{M}_{L\star P}$. The diagonal acts by the identity, so its module is the regular identity bimodule. These are the evaluation maps themselves and have their associative and unit cells.

The same construction applies to the mixed-model generated words: the graph and rectangle control formulas in Proposition 6.3 were stated for $M \preceq N \preceq L$, not only for one model. Thus all their actual kernel maps are retained before localization. A refinement graph acts by F_u, an equivalence on the profile categories by Theorem A. The universal localization at these specified arrows therefore admits the Morita functor. This is localization at actual refinement arrows, not a quotient of kernels by equality of actions. To view the target on the canonical finest model, replace the action of $P : (M, S) \to (N, T)$ by $F_{r_N} \Phi_P G_{r_M}$. The actual adjunction units and counits identify compositions and send each refinement graph to the identity equivalence. Theorem A supplies independence of common refinement. The specified natural transformation induced by α_P in Theorem B identifies the one-model coarsening squares at the level of actual kernels first, and hence at this Morita realization; it is not deduced backwards from an equivalence of actions.

For completeness the supported action is an actual geometric formula. Let D denote geometric Verdier duality and label the corresponding supported kernel by $P^D = D_{S_M \times T_N} P$, retaining its contragredient actions. It is left H_a^{op} and right H_b^{op}, in the symmetry convention fixed above; the geometric endpoints have not been exchanged. For a supported coefficient N_0, view $p_S^! N_0$ as a left $p_S^* H_a$-module using the actual map $p_S^* H_a \otimes p_S^! N_0 \to p_S^!(H_a \otimes N_0) \to p_S^! N_0$. The Hom below is Hom of left modules. Precomposition with the right H_b-action on P supplies the left H_b-action on its value, equivalently the required right H_b^{op}-action. Set

$$\Psi_{P^D}(N_0) = Rp_{T,*} R\mathcal{H}om_{p_S^* H_a}(P, p_S^! N_0).$$

If $N_0 = D_S M_0$, the bounded integrand established above and actual support duality give

$$\begin{aligned} D_T \Phi_P(M_0) &\simeq Rp_{T,*} D_{S_M \times T_N}(p_S^* M_0 \otimes^L_{p_S^* H_a} P) \\ &\simeq Rp_{T,*} R\mathcal{H}om_{p_S^* H_a}(P, p_S^! D_S M_0). \end{aligned}$$

Put $\omega = \omega_{S_M \times T_N}$. The middle identification is the actual balanced tensor–Hom adjunction $R\mathcal{H}om_Q(p_S^* M_0 \otimes^L_{p_S^* H_a} P, \omega) \simeq R\mathcal{H}om_{p_S^* H_a}(P, R\mathcal{H}om_Q(p_S^* M_0, \omega))$. It holds in the ambient module category by the relative bar/Hom adjunction; no interchange of an infinite limit with a colimit is used. The bounded integrand identifies its left side with geometric Verdier duality, and actual exchange for p_S identifies the inner Hom on the right with $p_S^! D_S M_0$. This proves the boundedness of this particular Hom integrand as well as the displayed comparison; it does not assert boundedness of arbitrary relative internal Hom. No tensor-dual formula for an arbitrary attached coefficient is used. The pairing is evaluation followed by the geometric support trace. It specifies the comparison, shows it is natural in coefficients and kernels, and proves that this already defined Hom expression is bounded and lies in the supported profile type. Biduality gives every supported input in this form. Applying the same evaluation on a threefold product and transitivity of trace identifies its convolution mate with $D(L \star P)$, and dualizes the actual localization boundary. The coefficient refinements on this side are F_u^D, as required. This proves Verdier compatibility of the model comparisons with actual geometric mates, rather than only a formal opposite action. □

Remark 6.6 (Endpoint scope of the kernel theorem). The one-sided counit theorem uses the same actual model change on both endpoints of a one-model kernel. Mixed-model generated words are realized geometrically and refinement graphs are then localized. Section 7 extends the rectangular part to independent presentations and gives a counterexample for unrestricted coupled kernels. We do not assert that all constructible kernels belong to the generated source. A kernel category need not be unchanged as a category of actual kernels even when its Morita realization is compatible with changes of presentation.

7 Kernels between curve presentations

7.1 Rectangles and independent model changes

The two endpoints need not belong to the same curve for rectangular kernels. The distinction between these kernels and geometrically coupled kernels is essential, even when their actions on selected coefficients agree.

Fix a finite labelled family $(\mathscr{M}_i, \mathfrak{p}_i)_{i\in I}$ of integer-cut components and geometric profiles over the same trait and residue field. Choose one presentation M_i for each label. Objects are the pairs (i, S) in their finite support diagrams. Define a geometric kernel source $\mathcal{K}(\mathbf{M})$ by the following generators: within each label, all same-model generators of Definition 6.1; between any two labels, all actual rectangles $C \boxtimes_Q E$ with coefficients in their independently defined profile categories. Take finite convolutions, cones, shifts and retracts, with full actual kernel mapping spectra. No comparison between the generic curves, no common coarse model, and no invertibility condition on a kernel counit enters this definition. Cross-label isomorphism graphs and general correspondences are not additional generators, even if two labels happen to denote isomorphic curves.

Theorem 7.1 (Independent model changes for the rectangular extension)**.** *For independently chosen actual modifications $u_i : M_i' \to M_i$ of these presentations, the actual kernel functors*

$$T_{ij}(P) = V_j \star P \star U_i \simeq R(u_{i,S} \times u_{j,T})_* P$$

define a unital, strongly convolution-compatible functor $\mathcal{K}(\mathbf{M}') \to \mathcal{K}(\mathbf{M})$. For every generated $P : (i, S) \to (j, T)$ the actual target counit

$$U_j \star T_{ij}(P) \longrightarrow P \star U_i$$

is an equivalence of kernels. All generated kernels, action integrands and convolution integrands are bounded constructible. The multi-object Morita realization and its actual Verdier mates are compatible with these comparisons and coherent strings of independent model changes. At the level of this realization the result is independent of the chosen presentations in each component. It does not assert an equivalence of the actual kernel categories.

Proof. Within one label the controlling kernels are the support graphs $G_i(W)$ and rectangles of Proposition 6.3. Add all cross-label rectangles $R_{ij}(C, E)$ to that controlling family. A pair of graph kernels can compose only within the same label, where its product is the graph of the support intersection. Composing a graph with a rectangle, on either side, restricts and extends by zero the corresponding coefficient. For two rectangles, possibly joining three different curves, the actual product formula is

$$R_{jk}(C', E') \star R_{ij}(C, E) \simeq R\Gamma_c(T_{M_j}, E \otimes^L_{H_j} C') \otimes_Q R_{ik}(C, E').$$

These are the same support and projection maps on the threefold product over k; they do not require the outer factors to be the same curve. Lemmas 5.4 and 6.2 prove preservation and finiteness. The four integrand checks in Proposition 6.3 likewise use only the intermediate algebra and the corresponding coefficients, so apply on these products of different curves. Finite stable closure gives boundedness and action preservation for the full generated source. In particular any word that crosses labels reduces to the finite stable envelope of rectangles between its two endpoints, even if it returns to its starting label. It creates no unexamined coupled graph.

For a rectangle on the fine presentations, proper product base change and the actual graph projection formulas give

$$T_{ij}(C' \boxtimes E') \simeq G_i C' \boxtimes G_j E',$$
$$U_j \star T_{ij}(C' \boxtimes E') \simeq G_i C' \boxtimes F_j G_j E',$$
$$(C' \boxtimes E') \star U_i \simeq G_i C' \boxtimes E'.$$

Under these specified kernel comparisons the actual map is $1 \boxtimes \epsilon^{j}_{E'}$. Theorem A for the jth curve makes it invertible. The modifications at i and j have not been identified. For a support graph, both endpoints have the same label, hence use the same u_i; Theorem B gives exactly the required actual counit. The controlling family and exactness in finite cones and retracts prove the counit for every generated kernel, naturally on all its actual maps. They also show that T_{ij} lands in the specified coarse source.

For $P : i \to j$ and $L : j \to k$, the composition comparison inserts this same counit:

$$V_k \star L \star U_j \star V_j \star P \star U_i \longrightarrow V_k \star L \star P \star U_i.$$

It is invertible by the assertion just proved for P. A diagonal unit has one label at both ends, so its unit comparison is the actual algebra unit for u_i, not an assertion about two unrelated contractions. Adjunction triangles give the unit and associative cells; the actual graph maps and product pushforwards give their compatibility with strings of modifications.

The action of each generated kernel preserves the appropriate profile category. Enriched Yoneda therefore gives a right-target- perfect multi-object Morita bimodule, and its tensor composition is the actual action composition. The geometric tensor–Hom and Verdier argument of Theorem 6.5 applies to $S_{M_i} \times_k T_{M_j}$ without change; the integrands needed there have just been proved bounded. Evaluation and trace specify the supported mates of the same comparisons. Finally each $\mathscr{M}_i$ has its own canonical finest presentation. Apply Theorem A independently in each component and conjugate the kernel actions by these coefficient equivalences. The actual counit squares just proved compare coarsening with that conjugation. Common-refinement independence and higher cells follow componentwise from A; no claim that all the curves have a common geometric model is made. □

7.2 The obstruction for unrestricted coupled kernels

Example 7.2 (An obstruction to arbitrary independent endpoint descent). Let $u : M' \to M$ be a one-cut modification of a thickness-two node q, and write Y', Y for the special fibres. Choose a profile in which the exceptional component is unmarked, so that both M and M' are presentations. Take two labelled copies of Y' with this profile. Leave the source presentation unchanged and contract the target by u. Consider the actual identity graph kernel $P = \Delta_* H'$ between the two copies. If one admitted this cross-label graph with arbitrary independent endpoint changes, its proposed comparison would be

$$U_u \star V_u \longrightarrow \Delta_* H'.$$

This map is not an equivalence of actual kernels.

Indeed choose two distinct geometric points x, y in the open exceptional component C°. Both map to q and have nearby algebra Q. In the graph convolution the intersecting support is $Y' \times_Y Y'$. Its projection to $Y' \times_k Y'$ is a closed immersion, since Y is separated, and the middle coordinate is uniquely determined. Pushforward on this support is therefore extension along that closed immersion. Taking stalks of the relative tensor in the ambient module category gives

$$(U_u \star V_u)_{(x,y)} \simeq Q \otimes^{L}_{A_q} Q, \qquad (\Delta_* H')_{(x,y)} = 0.$$

The restrictions of $u^{-1}i_q^*H$ and H' to C° are the constant algebra families A_q and Q. Choose an etale path between x and y on this connected stratum. Naturality of the actual algebra map along that path identifies the two stalk augmentations up to homotopy. Only existence of this identification, not a canonical choice or a claim that its space is contractible, is needed below. The balanced multiplication $Q \otimes^L_{A_q} Q \to Q$, together with its unit from Q, has composite id_Q. Consequently this stalk is nonzero. No boundedness or finite-dimensionality of the bare projection kernel is needed for the contradiction; failure of those properties would be a further obstruction to enlarging the source.

The failure persists despite equality of the induced actions on the selected profile category: Theorem A makes F_uG_u the identity there. Thus coefficient descent cannot prove this missing kernel comparison. The example does not contradict Theorem B, which contracts both endpoints by the same map, nor Theorem 7.1, whose cross-label source has independently specified rectangular generators.

7.3 Finite etale support spans

We next enlarge only the basic geometric kernels of Theorem B. The trait, geometric coefficients, integer-cut models, profiles and products over k are unchanged. The additional kernels are finite etale support spans. Their model changes must be Cartesian; this is an independently checkable geometric requirement, not a condition on their coefficient actions or their counits.

Fix a finite generic diagram $\mathcal{D}$ of finite etale covers and spans over the fixed generic base curve. A presentation consists of a base model X of a fixed profile $\mathfrak{p}$ and actual finite etale model covers $\pi_i : X_i \to X$, together with actual finite etale model spans for the specified generic spans, all with their generic identifications. This is a condition on actual schemes and maps, not on a coefficient descent functor. Finite disjoint unions of nodal models are allowed. Include identity spans and their support restrictions. Give each X_i the marking pulled back from $\mathfrak{p}$ on the canonical finest models. For each base support S, retain its full inverse image S_i in X_i. We use these finite inverse-image support diagrams, not arbitrary additional supports on the covers. Existence of such a presentation is not asserted for every proposed generic diagram on every model.

Lemma 7.3 (Finite etale maps preserve the lifted profile)**.** *The lifted markings define geometric profiles. Finite etale maps between these covers preserve the profile categories by actual pullback and finite direct image, with their supported Verdier mates. The same holds on their full inverse-image support restrictions. For an integer-cut modification $u : X' \to X$, the pullbacks $X'_i = X_i \times_X X'$ are actual integer-cut modifications of X_i, and the finest-model and coefficient comparisons are the actual etale and proper base-change comparisons.*

Proof. Formation of the reduced nonregular locus commutes with etale base change. The successive blow-ups in Lemma 5.1 therefore do so as well; their local node thicknesses and cuts are unchanged. Hence the finest model of X_i is the pullback of that of X. The generic-identity comparison is the unique one described in that lemma.

A finite etale cover of a closed unmarked block is a disjoint union of copies of that block. For a point this is immediate. For a rational chain, pull the cover back to its normalization. Each connected cover of a $\mathbf{P}^1$ component has degree one: the unramified Riemann–Hurwitz formula gives $2g - 2 = -2d$, so $d = 1$. At nodes an etale cover glues the same number of sheets of the two branches. The incidence graph is a tree, so successive choices at its leaves trivialize these sheet permutations without a cycle condition. Thus the whole block cover splits. In particular the unmarked pieces upstairs are again rational paths; no circular unmarked block is introduced. Marked components remain marked and may retain arbitrary horizontal monodromy.

Actual etale nearby base change identifies H_{X_i} with $\pi_i^* H_X$, including algebra actions and branch maps. Pullback preserves the stalk and lisse conditions. Finite etale direct image has stalks that are finite sums over sheets. It preserves lisse coefficients on marked strata, perfect modules at nodes, and the internal branch isomorphisms on each split block. There is no new condition on exterior attachments. These operations are bounded; no invariants or division by the degree is used. The support assertions follow by their Cartesian support squares. Finally $f^! = f^*$ for an etale map and $Rf_! = Rf_*$ for a finite map give the actual Verdier exchanges, evaluation and finite trace. Applying the same geometric base-change maps to u proves the last assertion. □

For supports S, T and a support $W \subseteq S \cap T$ in the base diagram, represent a specified span of $\mathcal{D}$ by a finite etale X-scheme Z with X-maps

$$X_i \xleftarrow{p} Z \xrightarrow{q} X_j.$$

Both maps are finite etale: etale locally on X they are maps of finite sets of sheets. Write Z_W for the full special-fibre inverse image of W, and form the actual kernel

$$K_{ij}(Z, W) = (p, q)_! H_{Z_W} \quad \text{on } S_i \times_k T_j.$$

The map factors as a finite map to the product over W, followed by the support immersion into the product over k. Both actions come from the actual etale identifications of nearby algebras. Include all these span kernels, and all rectangles of independently defined profile coefficients on the S_i, then take finite convolutions, cones, shifts and retracts with full actual kernel mapping spectra. Denote this source by $\mathcal{K}_{\mathrm{et},\mathcal{D}}(X)$. The identity spans and their support restrictions include the old units and support graphs; include their actual localization arrows. This definition does not refer to invertibility of any descent map.

Lemma 7.4 (Existing finite etale presentations form Cartesian squares)**.** *Let $u : X' \to X$ be an allowed base modification and suppose both models carry actual finite etale presentations of the same generic diagram $\mathcal{D}$. Their cover and span squares are canonically Cartesian, compatibly with composition. Conversely an existing presentation on X pulls back to one on X'.*

Proof. Nodal total spaces $xy = t^m$ are normal, as follows from the hypersurface S_2 property and regularity in codimension one. Finite etale covers are normal as well. The pullback $X_i \times_X X'$ and the independently given X_i' are finite normal covers of X' with the same identified generic cover; both are its normalization in that cover. Their unique isomorphism identifies the generic maps, hence the whole diagram, by separatedness and generic density. The same applies to every span model Z. Thus Cartesian compatibility is a proved consequence of these geometric presentations. Conversely any existing presentation pulls back to one on X'.

The uniqueness of these generic-identity comparisons also identifies their composites on any string of modifications. □

The fine kernel source is defined by its actual finite etale model generators and its selected coefficients, not as the image of a coarse kernel source. The comparison just proved permits us to write $Z' = Z \times_X X'$ in the proof. We do not assume that every arbitrary cover first given on X' has a presentation on every coarser model. The common base is part of the geometric cover diagram; its presentation is allowed to change, and is not a distinguished coarse anchor for descent.

Theorem 7.5 (Theorem B with finite etale support spans)**.** *Fix the generic finite etale cover/span diagram $\mathcal{D}$ and its prescribed profiles and support diagram. Let $u : X' \to X$ be an*

allowed actual base modification, with both X and X' carrying existing finite etale presentations of that same generic diagram. Their cover and span squares are canonically Cartesian by Lemma 7.4. At each such actual presentation, every generated kernel in $\mathcal{K}_{\mathrm{et},\mathcal{D}}(X)$ and each action or convolution integrand is bounded constructible. The resulting actions preserve the selected profile categories. For this u and every generated fine kernel $P : i \to j$, actual coarsening has values in the specified coarse source and satisfies

$$T(P) = V_j \star P \star U_i \simeq R(u_i \times u_j)_* P, \qquad U_j \star T(P) \xrightarrow{\sim} P \star U_i,$$

where the displayed map is the actual target counit. It preserves units and convolution strongly, and gives the corresponding actual Morita–Verdier comparisons. These are coherent on strings of Cartesian modifications and independent of common refinement among existing presentations of the fixed finite etale diagram. No existence of a presentation on an arbitrary coarser model is asserted.

Proof. Write $p_W : Z_W \to W_i$, $q_W : Z_W \to W_j$ for the finite etale maps and e for the indicated support inclusions. The span action is

$$\Phi_{K_{ij}(Z,W)} = e_{W_j,T_j,!}\, Rq_{W,*}\, p_W^*\, e_{W_i,S_i}^*.$$

It preserves the profile by Lemmas 7.3 and 5.4. Actual finite base change and regular-module cancellation give

$$K_{jk}(Z_2, V) \star K_{ij}(Z_1, W) \simeq K_{ik}(Z_1 \times_{X_j} Z_2, W \cap V).$$

The new middle scheme is finite etale over X, so it is a controlling span obtained by composing the specified generators. The formula includes multiplicities as actual sheets, not their Euler characteristics. In the first mixed formula use $p_W : Z_W \to W_i$ and $q_W : Z_W \to W_j$ as above. In the second use a separate span $X_j \leftarrow \widehat{Z} \to X_k$, with maps $\widehat{p}_V : \widehat{Z}_V \to V_j$ and $\widehat{q}_V : \widehat{Z}_V \to V_k$. Then

$$R_{jk}(C, E) \star K_{ij}(Z, W) \simeq R_{ik}\big(e_{W_i,S_i,!} Rp_{W,*} q_W^*(C|_{W_j}), E\big),$$
$$K_{jk}(\widehat{Z}, V) \star R_{ij}(C, E) \simeq R_{ik}\big(C, e_{V_k,T_k,!} R\widehat{q}_{V,*} \widehat{p}_V^*(E|_{V_j})\big).$$

Rectangle–rectangle multiplication is the formula in Theorem 7.1. All four are actual kernel projection and support base-change maps with both outer algebra actions.

These formulas give a controlling family of spans and rectangles. Before direct image, span–span integrands cancel the intermediate regular algebra on each finite sheet. Mixed integrands give the finite etale restrictions of the selected rectangle coefficient; the two-rectangle case uses the bounded relative tensor of selected coefficients. The finite maps have finite fibres, and support immersions preserve constructibility. Thus these integrands are bounded constructible before convolution. The same checks apply to a coefficient action. The four formulas, followed by exactness in each variable under finite cones and retracts, prove all the stated boundedness and preservation assertions for the generated source.

It remains to verify the kernel counit, not merely its action. For a span let $v : Z' \to Z$ be the Cartesian modification. Its nearby algebra unit is an equivalence by actual proper nearby base change, exactly as in Lemma 4.2:

$$H_Z \xrightarrow{\sim} Rv_* H_{Z'}.$$

Support restriction uses complete inverse images. Proper support base change consequently identifies $T(K_{ij}(Z', W'))$ with $K_{ij}(Z, W)$. For the target-side pullback, the square

$$\begin{array}{ccc} Z' & \xrightarrow{q'} & X'_j \\ v\downarrow & & \downarrow u_j \\ Z & \xrightarrow{q} & X_j \end{array}$$

is Cartesian by Lemma 7.4. Put $\kappa'(z') = (pv(z'), q'(z'))$ on $Z'_{W'}$, with target $S_i \times_k T'_j$, and write $v_W : Z'_{W'} \to Z_W$. The actual graph and support base-change maps identify

$$U_j \star T(K_{ij}(Z', W')) \simeq R\kappa'_!(F_{v_W} G_{v_W} H_{Z'_{W'}}),$$
$$K_{ij}(Z', W') \star U_i \simeq R\kappa'_! H_{Z'_{W'}}.$$

The first identification uses $q'^* H_{X'_j} \simeq H_{Z'}$ and regular cancellation; the second composes the source graph with (p', q'). Under these specified kernel comparisons the map is exactly

$$R\kappa'_!(\epsilon_{v_W, H_{Z'_{W'}}}).$$

The coefficient counit is the inverse of the pulled-back algebra unit by the adjunction triangle identity, also after the complete support restriction. Its pushforward is therefore an equivalence of kernels. This identifies the actual map, in addition to its two object values. On rectangles the map is $1 \boxtimes \epsilon_{E'}$, as before. The controlling family and finite stable closure prove the counit and preservation by T for every generated kernel.

Inserting this same middle counit proves strong convolution; the identity span has the actual algebra unit comparison. All maps are proper or etale base-change maps and adjunction units/counits. Their composition identities, together with the associators of actual fibre products, provide coherence on strings. The proved coefficient preservation gives the multi-object Morita realization. The bounded integrands allow the actual tensor–Hom/Verdier argument of Theorem 6.5; finite etale exchange identifies its additional graph mates and their traces.

Finally pull any existing diagram presentation to the canonical finest base model. Lemma 7.3 gives the finest cover models and the same lifted markings. For presentations with specified identical generic cover diagrams, their pullbacks agree: a finite etale cover of the normal finest base is the normalization in its generic finite cover, and hence is uniquely determined. The same holds for the span covers and their maps. This compares existing presentations, without asserting their existence on every coarser model. The preceding Cartesian comparisons and Theorem A therefore give common-refinement independence with their higher cells. A distinguished coarse anchor was not used to define the coefficient or span conditions. □

8 A cyclic curve and its kernels

8.1 Continuous holonomy and localization

Proposition 8.1 (A geometric cyclic test for Theorem A)**.** *There is an actual proper nodal curve model over* $\mathbf{C}[[t]]$ *with a profile having two incomparable integer-cut presentations, and a geometric nearby-module coefficient with nontrivial continuous loop holonomy. It has a nonzero localization boundary and admits a nonzero noninvertible localization kernel. Its coefficient, boundary and Verdier mate are preserved by Theorem 5.6.*

Proof. Take the projective cubic

$$M_0 : \quad xyz + t^3(x^3 + y^3 + z^3) = 0 \quad \subseteq \mathbf{P}^2_{\mathbf{C}[[t]]}.$$

It is flat and proper. Its generic fibre is smooth: a singular point cannot have a zero coordinate by the three partial derivatives; if all coordinates are nonzero, multiplying those derivatives gives $1 = -27t^9$, impossible in $\mathbf{C}((t))$. The special fibre Y_0 is the triangle of the three coordinate lines. At $[1 : 0 : 0]$ put $h = 1 + y^3 + z^3$, a unit, and $a = y$, $b = -z/h$. Then $ab = t^3$ and the coordinate map has invertible Jacobian at the node. The other two nodes are identical by symmetry. Thus all three nodes have thickness three in actual etale charts.

Two actual scheme presentations are provided by the blow-ups of

$$\mathcal{I}_1 = \mathrm{im}(\mathcal{O}_{M_0}(-1) \xrightarrow{y} \mathcal{O}_{M_0}) + (t), \qquad \mathcal{I}_2 = \mathrm{im}(\mathcal{O}_{M_0}(-1) \xrightarrow{y} \mathcal{O}_{M_0}) + (t^2).$$

Near either node on the line $y = 0$ these are exactly the cut ideals (a, t) and (a, t^2), giving cuts $\{1\}$ and $\{2\}$. Off those nodes the ideals are invertible: away from $y = 0$ they are the unit ideal, and at a smooth point of that component the cubic equation expresses y as a multiple of t^3 in a local trivialization. Hence these global projective blow-ups are the stated integer-cut modifications and are incomparable. The third node is unchanged. Their common finest refinement resolves all three thickness-three nodes. Its dual graph is a cycle with nine vertices. Mark the three valuations of the original coordinate lines. The six unmarked vertices form three two-component paths, so this is a profile as in Definition 5.2. Choose its finite closed-support diagram to be generated by the three closed unmarked chain blocks and the whole fibre. On M_0 these blocks are precisely its three nodes; on every finer presentation they are their complete inverse images. Thus the open U, each node support on M_0, and all localization triangles used below are actually included in the chosen support diagram.

We construct holonomy on the actual special curve Y_0, not on an abstract graph. Normalize its three lines to three copies of $\mathbf{P}^1$ and name their node preimages $0, \infty$ in cyclic order. For each positive integer N, glue $3N$ copies of $\mathbf{P}^1$ in a cycle and map them to the three original lines according to their index modulo three. This gives a finite morphism $Y_N \to Y_0$. On the smooth locus it is a disjoint union of identities; at every node it identifies the two branch parameters, hence induces an isomorphism of completed local rings $\mathbf{C}[[a, b]]/(ab)$. A finite map of these noetherian curves with that local completion is etale at the node. Rotation by three components makes it a $\mathbf{Z}/N$-torsor everywhere. The covers are compatible under divisibility of N.

Let $\lambda = 1 + \ell$ for odd ℓ and $\lambda = 1 + 4$ for $\ell = 2$. At precision r, use $N = \ell^r$ and the character $\mathbf{Z}/\ell^r \to (\mathbf{Z}/\ell^r)^\times$ sending 1 to λ. It is well defined since $\lambda^{\ell^r} = 1$ modulo ℓ^r. The associated sheaves and their compatible reductions give a rank-one continuous integral local system $L_{\lambda,\bullet}$ on Y_0. Its rationalization L_λ has holonomy $\lambda \neq 1$, and is constant on the smooth locus U, the disjoint union of three copies of $\mathbf{G}_m$. This construction exhibits both the actual covers and a stable lattice; continuity is not inferred from an arbitrary scalar label on a graph.

Put $F = H_{M_0} \otimes_Q L_\lambda$. It has module A_q at each node; its local vertical coefficient is the trivial unipotent coefficient, with the nontrivial holonomy in the horizontal gluing. Choosing trivializations on the normalization, arrange that a single chosen node q carries the nontrivial gluing factor. Its attachment is

$$A_q \longrightarrow B_0 \oplus B_\infty, \qquad w \longmapsto (\beta_0(w), \lambda\beta_\infty(w)).$$

In degree zero it sends 1 to $(1, \lambda)$, hence is nonzero. The actual mapping identification for localization proves that the q-summand of its boundary $i_* \bigoplus_{q'} A_{q'} \to j_! Q_U[1]$ is nonzero. Pullback identifies its fine coefficient with $H_{M'} \otimes_Q u^* L_\lambda$; inserting the exceptional components retains this holonomy and boundary by the proved full mapping comparison. The same applies to its Verdier mate. We assert this in the nearby-supported source; a global algebraic generic-fibre lift of L_λ is not needed or asserted here.

Finally let $E = j_! Q_U$, an H_{M_0}-module, and take the actual diagonal kernel $K = \Delta_* E$ with its two H_{M_0}-actions. Its action is $j_! j^*$, by the diagonal and open projection formulas. The same formulas give $K \star K \simeq K$ as actual kernels. It is nonzero, fixes $j_! Q_U$, and kills the nonzero object $i_* A_q$, so cannot be invertible. The complete inverse-image open U gives the corresponding kernel on either partial presentation with the same calculation. This verifies a noninvertible kernel and a convolution on the same geometric test. This supplies the concrete idempotent test for Theorem 6.4. □

8.2 A noninvertible rectangular kernel

Corollary 8.2 (A noninvertible rectangle on the cyclic test)**.** *On the actual curve of Proposition 8.1, let $j : U \hookrightarrow Y_0$ be the union of the three smooth open strata and put $E = j_! Q_U$. The actual bimodule kernel $P = E \boxtimes_Q E$ belongs to the geometric generated source and is nonzero and noninvertible. Its convolution is*

$$P \star P \simeq R\Gamma_c(U, Q) \otimes_Q P.$$

This kernel and its convolution descend from either of the two integer-cut presentations in that proposition by the actual comparisons of Theorem 6.4.

Proof. The coefficient E is in the profile category: it is constant on marked open strata and zero on all closed blocks. Its exterior attachments are permitted to be zero. On U the nearby algebra is Q_U, so extension by zero and the relative tensor projection formula give $E \otimes^L_H E \simeq E$. The rectangle calculation of Proposition 6.3 therefore gives, as an equivalence of actual kernels with both module actions,

$$(E \boxtimes E) \star (E \boxtimes E) \simeq R\Gamma_c(Y_0, E \otimes^L_H E) \otimes_Q (E \boxtimes E) \simeq R\Gamma_c(U, Q) \otimes_Q P.$$

It also gives the actual action $\Phi_P(M) = R\Gamma_c(U, j^*M) \otimes_Q E$. The restriction of P to $U \times U$ is the nonzero constant kernel. Moreover U is the disjoint union of three copies of $\mathbf{G}_m$, so $H^1_c(U, Q) = Q^{\oplus 3}$ and $H^2_c(U, Q) = Q(-1)^{\oplus 3}$; in particular its convolution above is nonzero. For any node q, the nonzero profile object $i_{q,*}A_q$ is killed by Φ_P. An invertible kernel would induce an invertible action, which cannot kill this object. Thus P is noninvertible. This argument does not infer a kernel equivalence from equality of actions: the displayed convolution was already computed in the actual kernel category.

For either modification $u : M' \to M_0$, the complete inverse-image open is still U. The ordinary support comparison identifies $F_u E$ with $E' = j'_! Q_U$. Form $P' = E' \boxtimes E'$ in that presentation. The actual rectangle pushforward and target counit of Theorem 6.4 give $T_u(P') \simeq P$ and $T_u(P' \star P') \simeq P \star P$. Hence this is a rectangular-kernel test on the same actual curve and presentations as the continuous holonomy and nonzero boundary, not only a test by modification equivalences or diagonal kernels. □

8.3 A finite etale quotient and coupled kernels

Example 8.3 (A coupled cyclic-cover kernel on the actual triangular curve)*.* Let N be the cubic model of Proposition 8.1 over $\mathbf{C}[[t]]$, and let $\sigma[x : y : z] = [y : z : x]$. The constant group $G = \langle \sigma \rangle \simeq \mathbf{Z}/3$ acts freely on N. Indeed the only projective fixed points of a nonidentity element are its three eigenlines. On each the cubic has value a nonzero scalar multiple of $1 + 3t^3$, a unit in the trait. Thus none lies on N, on either the generic or the special fibre. The permutation linearizes the ample bundle $\mathcal{O}_N(1)$. The projective finite-group quotient can therefore be constructed from its invariant section ring. Since the action is free and G is finite etale, the resulting map $p : N \to X = N/G$ is a finite etale G-torsor. Flatness, nodality and smoothness of the generic fibre descend along this etale cover; X is an actual proper nodal model. Its special fibre is the nodal one-gon quotient of the triangle, and its node has thickness three. Mark its original component on the finest model; the two unmarked components form a path returning to that marked component. Use the base support diagram generated by this closed block and the whole fibre, with its open complement. Take the cover family id_X, p and the generic graph of p and its transpose as the additional span generators. The lifted marking on N is the three-marked-component marking of Proposition 8.1. Here use the inverse-image support diagram: the base node pulls back to the union of all three node blocks, not to an individually selected one. This smaller diagram still contains the

full-node localization triangle used below; it need not contain every extra support of that proposition.

Now write p also for the special-fibre map. The graph kernel is $\Gamma_p = (\mathrm{id}, p)_* H_N$ and its transpose is $\Gamma_p^t = (p, \mathrm{id})_* H_N$ on the corresponding products over k; their actions are Rp_* and p^* respectively. Here transposition exchanges geometric endpoints; it is distinct from the supported Verdier dual, which keeps their order. The actual torsor identity $N \times_X N \simeq \coprod_{g \in G} N$, with its graph maps, gives on the special-fibre products

$$\Gamma_p^t \star \Gamma_p \simeq \bigoplus_{g \in G} \Gamma_g, \qquad \Gamma_p \star \Gamma_p^t \simeq \Delta_* R p_* H_N.$$

The latter coefficient is $H_X \otimes_Q p_* Q$; its permutation local system is retained, not replaced by a scalar degree. Moreover Γ_p itself is noninvertible. On the selected coefficient categories Rp_* has right adjoint p^*, and $p^* R p_* H_N \simeq H_N^{\oplus 3}$, which has rank three rather than one on each smooth stratum. Hence this adjoint composite is not the identity and Rp_* is not an equivalence; an invertible kernel would induce one. The action rotates components around the loop and supplies genuinely coupled kernels. For example Γ_σ on the product of two different smooth component strata is supported on its graph, so is nonzero and not lisse on that product. By Proposition 6.3, every original same-model kernel is in the finite stable envelope of diagonal support kernels and rectangles. The former vanish on this product and the latter are lisse there; finite cones and retracts remain lisse. Thus Γ_σ is not already in that original generated source.

Resolve the thickness-three node of X by the actual canonical integer-cut modification. Its pullback to N resolves the three nodes equivariantly. Theorem 7.5 applies to these Cartesian diagram presentations and preserves both displayed kernel convolutions and their Verdier mates. For the coefficient $F = H_N \otimes L_\lambda$ of Proposition 8.1, finite support base change identifies Rp_* of its localization triangle with that of $Rp_* F$. Its boundary remains nonzero: $p^* R p_* \delta_F \simeq \bigoplus_{g \in G} g^* \delta_F$ contains δ_F as a direct summand. Thus the coupled kernel also tests the actual nonzero supported boundary of the loop coefficient. This theorem does not permit contracting just one endpoint of a coupled graph as in Example 7.2.

9 Inputs and remaining scope

The fixed-node source and rational unipotent interface come from [1]; fixed-cut coefficient and support comparisons from [2]; and the actual one-sided kernel mechanism from [3]. The three direct input sources are retained in the local project archive `companion_sources/`, with byte fingerprints and original paths in `input_manifest.json`. The frozen companion sources and input manifest are retained locally and are not included in the public release. Their transitive companion snapshots are recorded separately and do not substitute for these direct inputs. The new profile, finest-model and consistent-image arguments prove Theorem 5.6; Theorems 6.4 and 6.5 prove the scoped kernel result. Theorem 7.1 extends it to independently presented curve families on the specified rectangular source; Example 7.2 rules out an unrestricted independent-endpoint version. Theorem 7.5 adds finite etale support spans under actual Cartesian changes of presentation, within the original curve and nearby-coefficient scope. Higher-dimensional toroidal modifications, relative trait-products of kernels and global completed-root exceptional operations remain separate problems.

Pinned input contract

All three direct manuscripts have source version 27 September 2026; the byte-exact copies were archived on 28 September 2026 as `Local_2026-09-27.tex`, `Refinement_2026-09-27.`

tex and Kernels_2026-09-27.tex. Labels below refer to those copies. The last column records the restriction used here, not a claim to import every conclusion of the companion.

Input and stable labels	Conclusion used	Hypotheses checked here
Local [1]: thm:frag-node-complete-local-datum; cor:frag-refinement-unipotent-interface	Actual node comparison; rational unipotent cochain-module, tensor and supported-duality interface.	Complete strictly henselian trait, ℓ invertible; no smooth disc factors; locally chosen coordinates and compatible base roots. Geometric nearby cycles carry no additional base-inertia representation. Coordinate changes are treated in Section 2; no global coordinate choice or restriction on the allowed horizontal monodromy is added. The local $\mathbf{Q}_\ell$ source has bounded finite-dimensional unipotent cohomology. General thickness uses the next input, not the thickness-one theorem alone.
Refinement [2]: prop: all-thickness-nearby; prop:cut-models	Primitive branch Kummer class at $xy = t^m$; actual integer-cut models and full resolution.	Fixed trait, positive integral thickness and actual integral cuts; geometric rational unipotent local source.
Refinement [2]: thm:cell-reconstruction	Chain reconstruction, including full mapping spectra.	Finite exceptional chains; unipotent smooth strata, perfect node values and invertible individual internal branch maps. This input reconstructs the chain internally. Arbitrary exterior attachments are compared here by the ambient adjunction in Theorem 4.3; no unit-generation claim on a global cyclic graph is imported.
Kernels [3]: lem:bounded-ambient; prop: ambient-kernel-types	Enhanced ambient module operations, bar/Fubini convolution and graph types; comparison with bounded objects. Public six-operation and adic inputs: [7, 8]; the Ind extension and bounded comparison are the companion's construction.	Rational constructible sheaves inside their Ind ambient category; right-adjoint values are compared only when bounded. Boundedness of our tensors and integrands is proved here, not inferred from the ambient formalism.
Kernels [3]: lem:support-intersection; lem:graph-adjunction	Actual support-intersection tensor identity and graph adjunction.	Locally closed immersions in the specified finite support diagram (including open, closed and mixed intersections); full inverse-image supports for model changes. Nearby-algebra maps and geometric graph operations are used. No unrestricted counit equivalence is assumed.
Kernels [3]: thm:one-sided-descent	One-sided counit and strong-convolution mechanism in the anchored source.	Its fixed local four-model domain is not identified with our profile domain. The mechanism is reused; source preservation, boundedness and the counit are verified anew above and for the stated rectangular and finite etale extensions.

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References

[1] Y. Jiang, *Local Kummer Realizations and Verdier Duality*, project manuscript, 27 September 2026. Local archive path: `companion_sources/Local_2026-09-27.tex`. Public current manuscript (not a replacement for the frozen version): https://github.com/weiruanj-code/AI4Math/tree/main/three_realization_submission.

[2] Y. Jiang, *Morita Coefficients, Incidence, and Semistable Refinement*, project manuscript, 27 September 2026. Local archive path: `companion_sources/Refinement_2026-09-27.tex`. Public current manuscript (not a replacement for the frozen version): https://github.com/weiruanj-code/AI4Math/tree/main/semistable_morita_refinement.

[3] *Verdier-Compatible Kernel Skeletonization under Anchored Semistable Refinement*, project manuscript, 27 September 2026. Local archive path: `companion_sources/Kernels_2026-09-27.tex`. Public current manuscript (not a replacement for the frozen version): https://github.com/weiruanj-code/AI4Math/tree/main/verdier_kernel_skeletonization.

[4] P. J. Haine, M. Porta and J.-B. Teyssier, *Exodromy beyond conicality*, arXiv:2401.12825v2, https://arxiv.org/abs/2401.12825v2, Theorem 0.3.1(3).

[5] J. Curry, *Dualities between Cellular Sheaves and Cosheaves*, arXiv:1607.06942, https://arxiv.org/abs/1607.06942, Theorem 6.12.

[6] M. Artin, A. Grothendieck and J.-L. Verdier (eds.), *Theorie des topos et cohomologie etale des schemas (SGA 4)*, Tome 3, Lecture Notes in Mathematics 305, Springer, 1973, Expose XII, Theorem 5.1.

[7] Y. Liu and W. Zheng, *Enhanced six operations and base change theorem for higher Artin stacks*, arXiv:1211.5948, https://arxiv.org/abs/1211.5948.

[8] Y. Liu and W. Zheng, *Enhanced adic formalism and perverse t-structures for higher Artin stacks*, arXiv:1404.1128, https://arxiv.org/abs/1404.1128.